

Functional ultrasound imaging of stroke in awake rats

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Abstract

Anesthesia is a major confounding factor in preclinical stroke research as stroke rarely occurs in sedated patients. Moreover, anesthesia affects both brain functions and the stroke outcome acting as neurotoxic or protective agent. So far, no approaches were well suited to induce stroke while imaging hemodynamics along with simultaneous large-scale recording of brain functions in awake animals. For this reason, the first critical hours following the stroke insult and associated functional alteration remain poorly understood. Here, we present a strategy to investigate both stroke hemodynamics and stroke-induced functional alterations without the confounding effect of anesthesia, i.e., under awake condition. Functional ultrasound (fUS) imaging was used to continuously monitor variations in cerebral blood volume (CBV) in +65 brain regions/hemisphere for up to 3hrs after stroke onset. The focal cortical ischemia was induced using a chemo-thrombotic agent suited for permanent middle cerebral artery occlusion in awake rats, and followed by ipsi- and contralesional whiskers stimulation to investigate on the dynamic of the thalamo-cortical functions. Early (0-3hrs) and delayed (day 5) fUS recording enabled to characterize the features of the ischemia (location, CBV loss), spreading depolarizations (occurrence, amplitude) and functional alteration of the somatosensory thalamo-cortical circuits. Post-stroke thalamo-cortical functions were affected not only early after the stroke onset but were also altered secondarily and remotely from the initial insult. Overall, our procedure enables early, continuous, and chronic evaluations of hemodynamics and brain functions which, combined to stroke or other pathologies, aims to better understand physiopathologies toward the development of clinically relevant therapeutic strategies.

eLife assessment

This **important** proof-of-concept study strongly supports the utility of functional ultrasound imaging for evaluating cerebral hemodynamics in rat models of brain injury. Functional ultrasound affords a distinct coverage/spatial/temporal resolution tradeoff when compared to other modalities for studying brain hemodynamics. The **solid** data presented indicate high fidelity of the recordings, a particular feat given that the rats were awake. On the other hand, single slice imaging and complexity of registration of subsequent imaging sessions limit the usefulness of the approach, particularly for quantitative imaging, and the small sample size will need to be followed up with and verified by future studies. This work will be of interest to researchers working in functional neuroimaging and more precisely with preclinical models of stroke in rodents.

Introduction

Stroke is a multifaceted and multiphasic pathology, complex to mimic under experimental conditions. Indeed, when compared to clinic, preclinical stroke models suffer from several limitations that narrow the experimental focus to few conditions^{1–3}. Among these limitations, one can highlight the complexity to combine (i) imaging stroke in conscious animal models, (ii) addressing post-stroke brain functions and (iii) recording of hyperacute stroke hemodynamics, all crucial to design timely effective therapeutic strategies.

As first limitation, the use of anesthesia in preclinical studies seems to hamper the transition from animal to patient as most of stroke occurs in awake or sleeping patients^{4,5}, but rarely in sedated patients. Moreover, anesthetics disrupt the brain functionality, alterates neurovascular coupling^{6,7}, while differentially affecting metabolism, electrophysiology, temperature, blood pressure and tissue outcome by acting as neurotoxic or neuroprotective agents (see reviews^{8–10}).

To date, only a few groups succeeded in inducing a stroke in awake rodents^{11–14}. Moreover, post-stroke network and functional alterations have been addressed by few preclinical studies, providing evidence of functional network reorganization from minutes^{15,16} to days^{17–20} following stroke onset. However, these studies mostly focused on the cortical readouts and were unable to capture how deeper brain regions, like thalamic relays, were functionally and/or temporally affected remotely from the stroke insult (e.g., diaschisis)^{21–23}. Furthermore, these studies were always conducted using various anesthetics (e.g., ventilated with halothane or isoflurane; medetomidine, urethane) known to differentially impact brain functions, as mentioned above.

Until recently, live monitoring of the hyperacute stroke-induced hemodynamics was restricted to few methods but often focused to the brain surface^{13,24,25}. On the other hand, functional ultrasound (fUS), a recent neuroimaging modality measuring cerebral blood volume changes (CBV)^{26–28}, was successfully employed to measure brain functions of awake rodents^{29–34}, to address early post-stroke functional reorganization¹⁶, and to track stroke-induced hemodynamics at the brain-wide scale (i.e., ischemia and spreading depolarization)³⁵. However, no study has further exploited such strategies to combine together stroke hemodynamics and brain-wide functional alteration in awake rodents.

In this study, we report on the stroke induction and the alteration of somatosensory brain functions in awake rats. Using the latest improvements toward imaging of awake rodents^{29,31,33} combined with chemo-thrombotic agent directly applied to the middle cerebral artery (MCA)^{36,37}, we were able to induce MCA occlusion (MCAo) in awake rats while capturing continuous hemodynamic changes, including ischemia and spreading depolarization, in +65 brain regions for up to 3hrs after stroke onset. Finally, we investigated on how somatosensory thalamo-cortical functional responses were progressively altered from early (0-3hrs) to late post-stroke (5d) timepoints.

Materials and methods

Animals

The experimental procedures were approved by the Committee on Animal Care of the Katholieke Universiteit Leuven, following the national guidelines on the use of laboratory animals and the European Union Directive for animal experiments (2010/63/EU). The manuscript was written according to the ARRIVE Essential 10 checklist for reporting animal experiments³⁸. Adult male Sprague-Dawley rats weighed between 250–400g (n=9; Janvier Labs, France) were used. During habituation rats were housed two per cage kept in a 12-hr dark/light cycle at 23°C with *ad libitum* access to water and controlled access to food (15g/rat/day). After the initial surgical procedure, rats were housed alone. See **Supplementary Table 1** reporting on animal use, experimentation, inclusion/exclusion criteria.

Body restraint and head fixation

The body restraint and head fixation procedures are adapted from published protocols and setup dedicated for brain imaging of awake rats^{39–41}. Rats were habituated to the workbench and to be restrained in a sling suit (Lomir Biomedical inc, Canada) by progressively increasing restraining periods from minutes (5mins, 10mins, 30mins) to hours (1 and 3hrs) for one or two weeks. The habituation to head-fixation started by short (5 to 30s) and gentle head-fixation of the headpost between fingers. The headpost was then secured between clamps for fixation periods progressively increased following the same procedure as with the sling. For both body restraint and head fixation, the initial struggling and vocalization diminished over sessions. Water and food gel (DietGel, ClearH2O, USA) were provided for all body restraint and head-fixation habituation sessions. Once habituated, the cranial window for imaging was performed as described below (**Figure 1A-C**).

Surgical procedures

Cranial window over the MCA: Rats were anesthetized with isoflurane (5% for induction, 2% for maintenance; Iso-Vet, 1000 mg/g, Dechra, Belgium) and fixed in a stereotaxic frame. The depth of anesthesia was confirmed by the absence of reflex during paw pinching. After scalp removal and tissue cleaning, a 1-mm² cranial window was performed at coordinates bregma +2mm and lateral 7mm, over the left distal branch of the MCA as reported in Brunner, Korostelev et al.¹⁶. A silicone plug (Body Double-Fast Set, Smooth-on, Inc., USA) was used to protect the window and ease the access to the MCA before the occlusion procedure. Then, a stainless-steel custom designed headpost was fixed with bone screws (19010-00, FST, Germany) and dental cement (Super-Bond C&B, Sun Medical Co., Japan) to the animal skull (**Figure 1B**, left) as previously described by Brunner, Grillet et al.³³.

Cranial window for imaging: After recovery and habituation to head-fixation, a second cranial window was performed between bregma -2 to -4mm and 6mm apart from the sagittal suture (same anesthesia settings as the first cranial window; see above) following the procedure

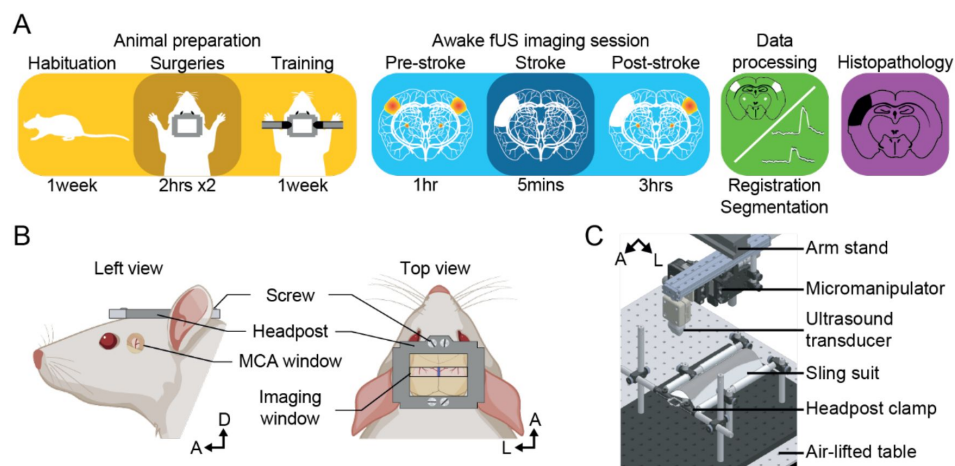


Figure 1.

Experimental procedure.

(A) Workflow for brain imaging of awake head-fixed rats including, from left to right: animal preparation (habituation to the bench, implantation of cranial windows, training), functional ultrasound (fUS) imaging of stroke induction and brain functions, data processing, and histopathology. **(B)** Overview of the headpost implantation and cranial windows developed for combined MCAo (left) and brain imaging (right) under awake conditions. **(C)** Computer-aided design of the experimental apparatus where the animal is placed and secured in a suspended sling suit and the head fixed by the means of clamps holding the headpost implanted to the rat skull. A, Anterior; D, Dorsal; L, Left.

described in Brunner, Grillet et al.,³⁴ (Figure 1B, right). This cranial window aims to cover bilateral thalamo-cortical circuits of the somatosensory whisker-to-barrel pathway. A silicone plug was also used to protect the window and a headshield was added to secure it.²⁹

For both cranial windows, the dura mater was kept intact. After each surgery, rats were placed in their home cage and monitored until they woke up. Rats were medicated with analgesic (Buprenorphine, 0.1mg/kg, Ceva, France), anti-inflammatory (Dexamethasone, 0.5mg/kg, Dechra, Belgium) drugs injected directly after the surgery, at 24hrs and 48hrs after the surgery. An antibiotic (Emdotrim, 5%, Ecuphar, The Netherlands) was added to the water bottle.

Positioning

The mechanical fixation of the head-post ensures an easy and repeatable positioning of the ultrasound probe across imaging session. The ultrasound probe is indeed fixed to a micromanipulator enabling light adjustments. To find the plane of interest (containing both S1BF and thalamic relays: bregma - 3.4mm), we used brain landmarks (e.g., surface of the brain, hippocampus, superior sagittal sinus, large vessels). Note that as the headpost was carefully placed in the same position relative to the skulls landmarks (bregma and lambda), the position of the region of interest was minimal across animals.

Chemo-thrombotic stroke induction with ferric chloride solution

Once body restrained and head-fixed the silicone plug covering the MCA window was removed allowing the application of a drop of 20% ferric chloride solution (FeCl_3 ; Sigma Aldrich, USA) to the MCA^{36,37} (Figure 2). Once the ischemia was visually detected using the real-time display of μ Doppler images, the solution was washed out with saline to stop the reaction.

Whisker stimulation paradigm

Two stimulation combs individually controlled by a stepper motor (RS Components, UK) were used to deliver mechanical 5-Hz sinusoidal deflection of $\sim 20^\circ$ of amplitude for 5s, alternatively to left and right whisker pads. For each whisker pad, trials were spaced by a period of 1min and 20s without stimulation. Thus, the effective delay between two stimulations delivered to the same whisker pad is 80 seconds from start to start. The blocks of stimulation were continuously delivered throughout the imaging sessions, time-locked with the fUS acquisition (Figure 3) to allow the subsequent analysis of hemodynamic responses within the fUS time-series.

Functional ultrasound imaging acquisition

Coronal μ Doppler images were acquired using a 15-MHz linear probe composed of 128 piezo-elements spaced by 100 μm (L22-14Vx, Vermon, France) connected to a dedicated ultrasound scanner (Vantage 128, Verasonics, USA) and controlled by a high-performance computing workstation (fUSI-2, AUTC, Estonia). This configuration allowed to image the brain vasculature with a resolution of 100 μm laterally, 110 μm in depth, and 300 μm in elevation.³⁴ The ultrasound sequence generated by the software is adapted from Macé et al.³¹ and Brunner, Grillet et al.³⁴ Ultrafast images of the brain were generated using 5 tilted plane-waves (-6° , -3° , $+0.5^\circ$, $+3^\circ$, $+6^\circ$). Each plane wave is repeated 6 times and the recorded echoes are averaged to increase the signal to noise ratio. The 5 plane-wave images are added to create compound images at a frame rate of 500Hz. To obtain a single vascular image we acquired a set of 250 compound images in 0.5s, an extra 0.3s pause is included between each image to have some processing time to display the images for real-time monitoring of the experiment. The set of 250 compound images has a mixed information of blood and tissue signal. To extract the blood signal we apply a low pass filter (cut off 15Hz) and an SVD filter that eliminates 20 singular values. This filter aims to select all the signal from blood moving with an axial velocity higher than $\sim 1\text{mm/s}$. To obtain a vascular image

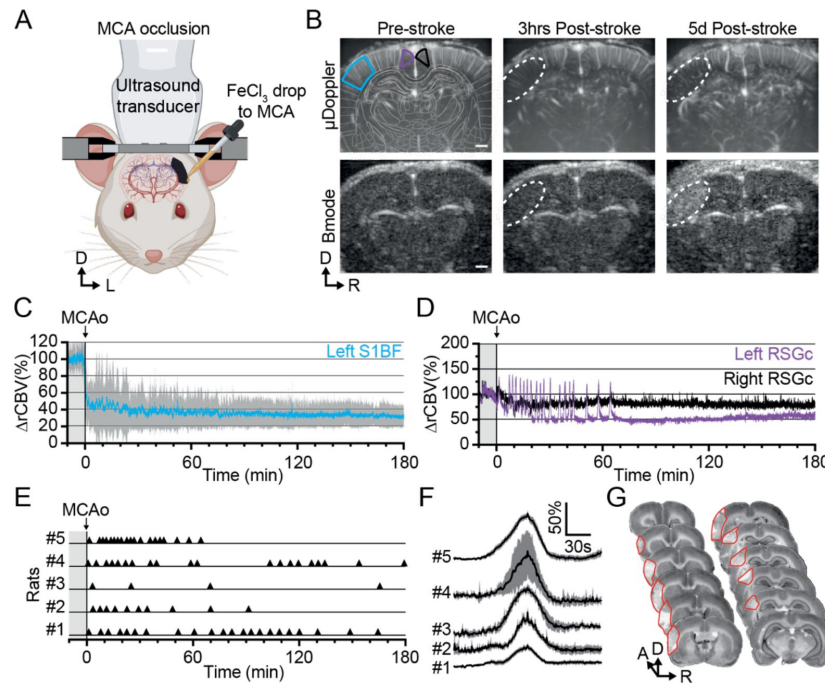


Figure 2.

FeCl₃-stroke induction under awake conditions.

(A) Front view representation of functional ultrasound (fUS) imaging during live chemo-thrombosis of the left MCA with FeCl₃ in awake head-fixed rats. **(B)** Set of typical coronal μ Doppler images of the brain microvasculature (top row) and morphological Bmode images (bottom row) before stroke (left), 3hrs (middle), and 5d after stroke onset (right) from the same animal. μ Doppler images (top left) were registered and segmented based on a digital version of the rat brain atlas (white outlines). Colored outlines (cyan, purple, and black) delineate regions of interest plotted in (C) and (D). The white dotted region of interest highlights the ischemia in μ Doppler images (Top row) and tissue hyper-echogenicity in Bmode (Bottom row). **(C)** Temporal plot of the average signal ($\Delta rCBV$ (%), mean $\pm$ 95%CI, n=5) in the barrel-field primary somatosensory cortex (S1BF, cyan) from the left hemisphere, affected by the MCAo. **(D)** Temporal plots of the average signal ($\Delta rCBV$ (%)) in the retrosplenial granular cortex (RSGc) from the affected (purple) and non-affected hemisphere (black) from the same animal. **(E)** Occurrence of spreading depolarizations after MCAo. Each horizontal line represents one rat; each triangle marker depicts the occurrence of one spreading depolarization. **(F)** Temporal plots of the average signal change ($\Delta rCBV$ (%), mean $\pm$ 95%CI, respectively black line and gray band) of hemodynamic events associated with spreading depolarizations (centered on the peak) for each rat (#1 to 5). **(G)** Typical rat brain cross-sections stained by cresyl violet to evaluate the tissue infarction at 24hrs after FeCl₃-induction occlusion of MCA. The infarcted territory is delineated in red. Scale bars: 1mm. D: Dorsal; L: left; R: right.

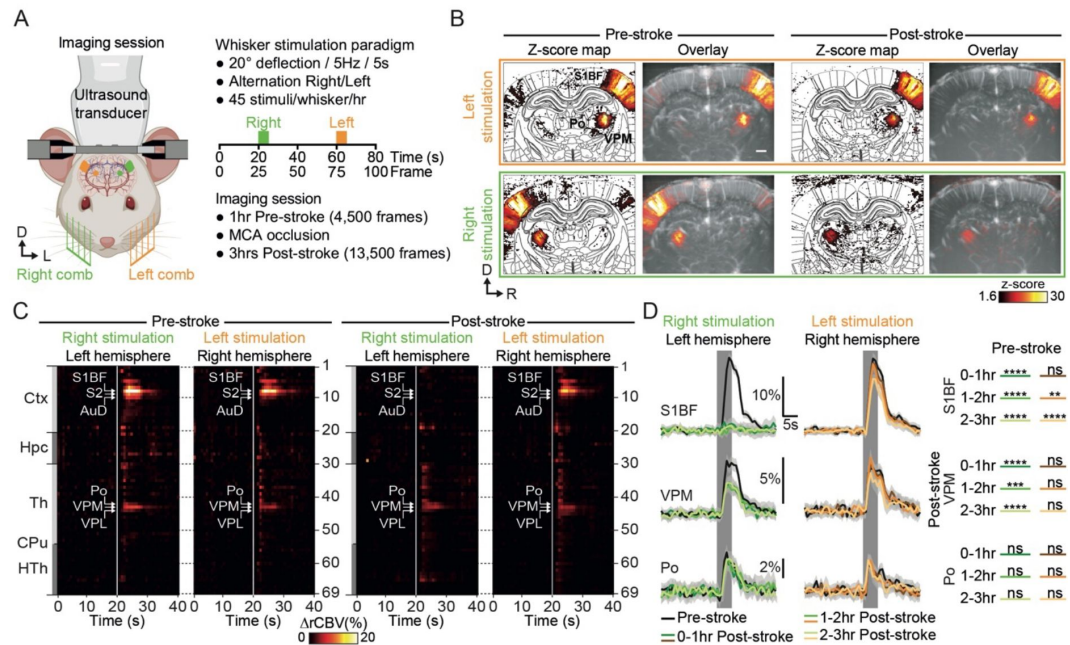


Figure 3.

Early post-stroke alteration of whisker-to-barrel thalamo-cortical circuit.

(A) Front view representation of functional ultrasound (fUS) imaging during repetitive stimulation of the left (orange) or right whisker pad (green) with a mechanical comb in awake head-fixed rats. Whisker stimulations were delivered alternately between left and right whisker pads before and early after MCAo. Each rat receives 45 stimuli per whisker pad each hour of imaging. (B) Average activity maps (z-score) from one rat depicting evoked functional responses to either left (orange) or right whisker pads stimulation (green) registered with a digital version of the rat Paxinos atlas (white outlines) and overlaid with the corresponding coronal μ Doppler image, before (left; Pre-stroke, average of 45 trials) and after stroke induction in the left hemisphere (right; Post-stroke, average of 125 trials). (C) Region-time traces of the average hemodynamic changes ($\Delta rCBV(\%)$) in response to right (green) or left whisker stimulation (orange) extracted from the contralateral hemisphere (left and right, respectively) before (left; Pre-stroke, $n=5$, 45 trials/rat) and after stroke induction in the left hemisphere (right; Post-stroke, $n=5$, 135 trials/rat). Brain regions are ordered by major anatomical structures (see [Supplementary Table 2](#)). The vertical line represents the stimulus start. S1BF, S2, AuD, VPM, VPL and Po regions are brain regions significantly activated (all p value <0.01 ; GLM followed by t-test). A larger version of panel C is provided in [Supplementary Figure 3](#). (D) Left, Average response curves from the S1BF, the VPM, and Po regions before (Pre-stroke, black, $n=5$, 45 trials/rat), and from first to third hour after stroke induction (0-1hr, 1-2hr, 2-3hr Post-stroke, orange and green, $n=5$, 45 trials/hr/rat). Data are mean $\pm$ 95%CI. The vertical bar represents the whisker stimulus. Right, Statistical comparison of the AUC between pre-stroke and post-stroke response curves for S1BF, VPM, and Po regions (Non-parametric Kruskal-Wallis test corrected with Dunn's test for multiple comparisons; ns: non-significant; * $p<0.05$; ** $p<0.01$; *** $p<0.001$; **** $p<0.0001$. See also [Supplementary Figure 4](#)). Scale bars: 1mm. D: Dorsal; L: left; R: right; Ctx: Cortex; Hpc: Hippocampus; Th: Thalamus; CPu: Caudate Putamen; HTTh: Hypothalamus; S1BF: barrel-field primary somatosensory cortex; S2: Secondary somatosensory cortex; AuD: Dorsal auditory cortex; VPM: Ventral postero-medial nucleus of the thalamus; VPL: Ventral postero-lateral nucleus of the thalamus; Po: Posterior nucleus of the thalamus.

we compute the intensity of the blood signal i.e., Power Doppler image. This image is in first approximation proportional to the cerebral blood volume^{26,28}. Overall, this process enables a continuous acquisition of power Doppler images at a frame rate of 1.25Hz during several hours. Then, the acquired images are processed with a dedicated GPU architecture, displayed in real-time for data visualization, and stored for subsequent off-line analysis³⁴.

fUS data processing and analysis

The data processing was performed following the procedure described by Brunner, Grillet et al.,³⁴.

Registration to Paxinos rat brain atlas and data segmentation

We registered the fUS dataset to a custom digital rat brain atlas used in Brunner et al.³⁵, using one coronal plane (bregma -3.4mm) from the stereotaxic atlas of Paxinos⁴². The image of the brain vasculature was manually translated and rotated to align it the coronal plane of the reference atlas. For an accurate registration, we used landmarks such as the surface of the brain, hippocampus, superior sagittal sinus, and other large vessels. If needed, the brain volume was scaled to fit the atlas outline. The outcome of this registration procedure is an affine coordinate transformation : $\vec{r}' = M\vec{r} + \vec{a}$, where $\vec{r} = (x, y, z)$ are the original coordinates image of the brain vasculature, M is the rotation and scaling matrix and $\vec{a}$ the translation vector. The dataset was segmented into 69 anatomical regions/hemispheres of the reference atlas (see **Supplementary Table 2**). The hemodynamic signals were averaged in each area. The segmentation and the data processing were performed using an automated MatLab-based pipeline. The software for data registration and segmentation is available in open-access³⁴.

Relative cerebral blood volume (rCBV)

We used the relative cerebral blood volume (rCBV, expressed in % as compared to baseline) to analyze ischemia, transient hemodynamic events associated with spreading depolarizations (SDs) and functional changes. rCBV is defined as the signal in each voxel compared to its average level during the baseline period. After registration and segmentation, the rCBV signal was averaged in each individual region.

Analysis of stroke hemodynamics

The extraction of the temporal traces from the ischemic area was performed based on the temporal analysis of the rCBV signal in the primary somatosensory barrel-field cortex (S1BF). The detection of hemodynamic events associated with SDs was performed based on the temporal analysis of the rCBV signal in the retrosplenial granular (RSGc) and dysgranular (RSD) cortices of the left hemisphere (ipsilesional). Hemodynamic events associated with SDs were defined as transient increase of rCBV signal (+25%) detected with a temporal delay of <10 frames (i.e., 8s) between the two regions of interest, validating both the hyperemia and spreading features of hemodynamic events associated with spreading depolarizations^{35,43-45}. This procedure allowed to measure the occurrence of hemodynamic event associated with SDs over the recording period. Live recording of ischemia and spreading depolarizations can be visualized in **Movie 1**.

Activity maps

Pre- and post-stroke recordings are reshaped in 40-s sessions, i.e., 50 frames, centered on the start of the stimulation (at 20s), and averaged based on the whisker stimulation paradigm (left or right). In each voxel, we compared signals along the recording in a time window before the stimulus onset and a time window after stimulus onset using two-tailed Wilcoxon rank sum test. We obtained the z-statistics of the test for each voxel, and consequently a z-score for the coronal cross-section. Mean activity maps for left or right whisker stimulation (**Figure 3B** and **4A**) show z-

score value calculated using a Fisher's transform for all voxels across the coronal cross-section. Only voxels with a z-score > 1.6 were considered significantly activated ($p < 0.05$ for a one-tailed test).

Hemodynamic response time-courses

The relative hemodynamic time course $\Delta rCBV$ was computed for each brain regions (after registration and segmentation; **Figure 3C-D** and **4B**), as the rCBV change compared to baseline at each time point. No additional filtering was used, and no trial was removed from the analysis.

Statistical analysis

Activated brain regions were detected from hemodynamic response time-courses using GLM followed by t-test across animals as proposed in Brunner, Grillet et al.³⁴. The area under the curve (AUC) from hemodynamic response time-courses was computed for individual trials in S1BF, VPM and Po regions, for all the periods of the recording and for all rats included in this work. AUC were compared and analysed using a non-parametric Kruskal-Wallis test corrected for multiple comparison using a Dunn's test. Tests were performed using GraphPad Prism 10.0.1.

Histopathology

Rats were killed 24hrs after the occlusion for histological analysis of the infarcted tissue. Rats received a lethal injection of pentobarbital (100mg/kg i.p. Doletal, Vetoquinol, France). Using a peristaltic pump, they were transcardially perfused with phosphate-buffered saline followed by 4% paraformaldehyde (Sigma-Aldrich, USA). Brains were collected and post-fixed overnight. 50- μ m thick coronal brain sections across the MCA territory were sliced on a vibratome (VT1000S, Leica Microsystems, Germany) and analyzed using the cresyl violet (Electron Microscopy Sciences, USA) staining procedure (see [Open Lab Book](#) for procedure). Slices were mounted with DPX mounting medium (Sigma-Aldrich, USA) and scanned using a bright-field microscope.

Results

Animals

Report on animal use, experimentation, exclusion criteria can be found in **Supplementary Table 1**. Rat #1 was excluded after the control session as the imaging window was too anterior to capture both cortical and thalamic responses. Rat #2 was excluded as hemodynamic responses were inconsistent during baseline (pre-stroke) period. Rat #9 showed early post-stroke reperfusion and was excluded from stroke analysis, the control session (pre-stroke) from Rat #9 was analyzed.

Real-time imaging of stroke induction in awake rats

We first developed a dedicated procedure for real-time imaging of stroke induction and associated evoked functional deficits in awake head-fixed rats (**Figure 1A**). Each rat was subjected to two cranial windows accessing independently the distal branch of the left MCA (**Figure 1B**, Left) and the selected brain regions to image (**Figure 1B**, Right). The latter was performed between bregma -2 and -4mm allowing for jointly monitoring the bilateral thalamo-cortical circuits of the somatosensory whisker-to-barrel pathway, including the ventroposterior medial nucleus of the thalamus (VPM) and the primary somatosensory barrel-field cortex (S1BF). Moreover, the selected coronal cross-section includes the posterior nucleus of the thalamus (Po), the reticular nucleus of the thalamus, and the ventral part of the zona incerta known for relaying information related to whiskers^{46,47}, and also direct efferent projections from the S1BF to other cortical and subcortical regions⁴⁸. Prior to imaging sessions, rats were extensively trained to accept

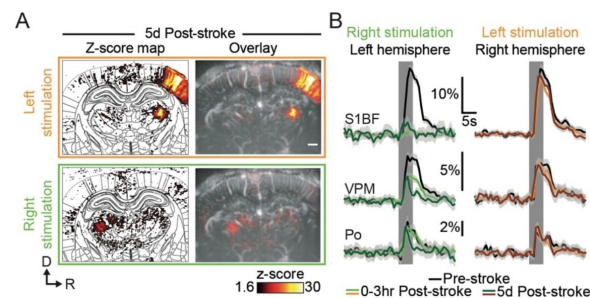


Figure 4.

Late post-stroke alteration of whisker-to-barrel thalamo-cortical circuit.

(A) Activity maps (z-score; average of 45 trials) depicting evoked functional responses to left (orange) or right whisker pads stimulation (green) 5d after stroke induction. Z-score maps are registered with the Paxinos atlas (white outlines; Left) and overlaid with the corresponding coronal μ Doppler image (Right). **(B)** Left; Average response curves to left and right whisker stimulation (orange and green; respectively) extracted from S1BF, VPM, and Po before (Pre-stroke, black, $n=2$, 45 trials/rat), 0-3hr (0-3hr Post-Stroke; light orange/green, $n=2$, 45 trials/hr/rat) and 5d after stroke induction (5d Post-stroke, dark orange/green, $n=2$, 45 trials/rat). Data are mean $\pm$ 95%CI. The vertical bar represents the whisker stimulus. Scale bars: 1mm. D: Dorsal; R: right; S1BF: barrel-field primary somatosensory cortex; VPM: Ventral postero-medial nucleus of the thalamus; Po: Posterior nucleus of the thalamus.

comfortable restraints in the experimental apparatus (**Figure 1C**), suitable for fUS recording of brain functions and stroke induction under awake conditions. After data acquisition, the coronal cross-section was registered and segmented on a custom-developed digital rat atlas⁴⁹ to provide a dynamic view of the changes in perfusion induced either by the stroke or evoked activity.

To overcome the limitations of conventional stroke models, we occluded the distal branch of the MCA by the mean of a chemo-thrombotic ferric chloride solution (FeCl_3)^{36,37} while performing fUS imaging in awake rats (**Figure 2A**). It should be mentioned that the rats did not show any obvious signs of pain or discomfort (e.g., vocalization, aggressiveness) during the restrain period and occlusion procedure. The MCA occlusion (MCAo) was captured live with fUS and confirmed by the large drop of signal, i.e., ischemia, localized in the cortex of the left hemisphere (**Figure 2B, C**, **Movie 1** and **Supplementary Figure 1**) as shown with $\mu\text{Doppler}$ image taken 3hrs and 5d after the stroke onset (dashed outline, **Figure 2B**, Top row). Bmode images accounting for the brain tissue echogenicity remain unchanged early after stroke onset (3hrs) while showing focal hyper-echogenicity (dashed outline, **Figure 2B**, Bottom row) lately after stroke onset (5d) as a marker of focal lesion⁵⁰. The stroke-induced hemodynamic changes have been continuously recorded for up to 3hrs after stroke onset, registered and segmented into 69 regions (**Supplementary Figure 1**). We first extracted the average change in rCBV (ΔrCBV in %) in the S1BF cortex of the left hemisphere (blue region, **Figure 2B**) and detected an abrupt drop of rCBV down to ~40% of the baseline level after the occlusion of the MCA, followed by a progressive decrease of the rCBV to 30% of baseline level 3hrs after the stroke onset (**Figure 2C** and **Movie 1**). Second, we extracted the average rCBV change from a cortical region supplied by the anterior cerebral artery directly after the MCAo. The signal extracted from the retrosplenial granular cortex (RSGc; purple and black regions in **Figure 2B**) shows successive and transient increases of signal. It characterizes hemodynamic events associated with spreading depolarizations (SDs) in the left hemisphere (in purple; **Figure 2D** and **Movie 1**) while resulting in a slight and stable oligemia in the right hemisphere (in black; **Figure 2D** and **Supplementary Figure 1**). SD events were observed in the peri-ischemic territory of all rats subjected to MCAo and occurred in an ostensibly random fashion (**Figure 2E**); however, hemodynamic events associated with SDs showed a similar bell shape and time-course across animals (**Figure 2F**). On average, we detected 5 SD events per hour per rat. Finally, we stained brain slices 24hrs after MCAo and confirmed that FeCl_3 -induced ischemia turned into tissue infarction (red delineation; **Figure 2G**).

Stroke-induced alterations of the thalamo-cortical functions

One hour before and during 3hrs after the occlusion of the MCA, rats received mechanical stimulation of the whisker alternately delivered to the left and right pad using motorized combs (5-Hz sinusoidal deflection, 20° amplitude, 5-s duration; **Figure 3A**) to capture the spatiotemporal dynamics of the functional circuit. Before stroke, the sensory-evoked stimulations elicited a robust and statistically significant functional response ($z\text{-score} > 1.6$, see Material and Methods) for both left and right stimulation (orange and green respectively; $z\text{-score}$ maps; Pre-stroke panel, **Figure 3B** and **Movie 2**) with the activity spatially confined in the contralateral dorsal part of the VPM and S1BF. The temporal analysis of the somatosensory evoked responses in the contralateral hemisphere confirmed that VPM, Po, and S1BF regions were significantly activated and for both left and right stimuli ($****p < 0.0001$, $***p < 0.001$ and $****p < 0.0001$ respectively; Left panel, **Figure 3C**). We also detected significant increase of activity in S2, AuD, Ect ($****p < 0.0001$) and PRh ($***p < 0.001$) cortices and VPL nucleus ($**p < 0.01$; the list of acronyms is provided in **Supplementary Table 2**), brain regions receiving direct efferent projections from the S1BF^{48,51,52}, VPM or Po nuclei^{53–55}. It is worth noted that no habituation or sensitization due to the repetitiveness of whiskers stimulation was observed in cortical and subcortical regions over the pre-stroke sessions (**Supplementary Figure 2**).

After the stroke, the activity map from the left pad stimulation elicited a similar response pattern as pre-stroke; however, the right pad stimulation showed a total absence of functional response in the S1BF cortex and a significant reduction of the response in the VPM (z-score maps; Post-stroke panel, **Figure 3B** and **Movie 2**). Over the 3hrs following stroke onset, functional responses to left whisker stimulation were still detected in the cortical and thalamic regions of the contralateral (right) hemisphere; however, functional responses to right whisker stimulation were only detected in subcortical nuclei (i.e., VPM, Po, VPL), while attenuated when compared with the responses from the pre-stroke period and from the other hemisphere (**Figure 3B, C**). Furthermore, no responses were detected at the cortical level (S1BF, S2, and AuD; right panel, **Figure 3B, C**). A larger version of **Figure 3C** is provided in **Supplementary Figure 3**.

To better evaluate how the functional responses were affected by the stroke, we have divided the post-stroke recording period into 3 sections of 1hr each and compared them with the 1-hr pre-stroke period (**Figure 3D**). Temporal plots from the pre-stroke period showed robust increases in signal during the stimulus in S1BF, VPM, and Po regions and high consistency between left and right stimuli (black plots, **Figure 3D** and **Supplementary Figures 2-3**); fitting well the hemodynamic response functions as previously observed^{16,56}. Indeed, the hemodynamic responses were characterized by a quick increase in signal during whisker stimulation reaching a peak after 4.0s at $18.2 \pm 1.3\%$ (4.0s, $18.6 \pm 1.2\%$) of baseline level for S1BF, 4.0s at $4.6 \pm 0.5\%$ (3.2s, $5.8 \pm 0.7\%$) for VPM, and 2.4s at $2.9 \pm 0.7\%$ (3.2s, $4.0 \pm 0.8\%$) for Po from the left stimulation (right, respectively; mean $\pm 95\%$ CI) before slowly returning to baseline level (black plots, **Figure 3D**).

During the first hour following the stroke onset, functional responses in the left hemisphere (i.e., ipsilesional) were abolished in the S1BF, S2, and AuD (0-1hr Post-stroke, ****pvalue<0.0001), significantly decreased in the VPM (0-1hr Post-stroke, ***pvalue<0.001), and unchanged in Po and VPL (0-1hr Post-stroke, ns; **Figure 3D**) when compared with the pre-stroke responses (Pre-stroke, black plots, **Figure 3D**). Over the two following hours (i.e., 1-2hr and 2-3hr Post-stroke), the hemodynamic responses captured in these regions remained similar as those detected during the first post-stroke hour (green plots, **Figure 3D**).

Regarding the right hemisphere (i.e., contralesional), the functional responses of S1BF and VPM were conserved during the first hour after the stroke onset (ns, 0-1hr Post-stroke; orange plots, **Figure 3D**). During the two following hours, signal changes in S1BF show a significant and progressive decrease of activity (1-2hr Post-stroke **pvalue<0.01, 2-3hr Post-stroke ****pvalue<0.0001; orange plots, **Figure 3D**); Similar observation were made for S2 and AuD) whereas responses in VPM remained stable during the second hour post-stroke (1-2hr, ns) before significantly decreasing during the third hour (2-3hr Post-stroke *pvalue<0.05; orange plots, **Figure 3D**). Finally, the functional responses in VPM and Po remained unchanged over the 3hrs following the stroke onset (bottom panel, **Figure 3D**).

Activity maps, region-time traces of the 69 brain regions, mean and individual time-course for all trials (left and right stimuli - including ipsi and contralateral traces), imaging timepoints (Control, Pre-Stroke, Post-Stroke) for all the rats included in this work can be found in **Supplementary Figure 5**.

Delayed alteration of the somatosensory thalamo-cortical pathway

A secondary objective of this work was to evaluate the fUS ability to identify potential delayed functional alteration within a few days after the initial injury. Two animals were imaged five days after the MCAo following the same experimental, stimulation, imaging, and processing conditions as for the early post-stroke session. As the number of rats imaged at this timepoint is limited to

$n=2$, we did not performed statistical analysis. Activity maps, region-time traces and individual trials for both right and lieft stimulation (including ipsi and contralateral) for the each rats are provided in **Supplementary Figure 5** [↗](#).

Five days after the MCA occlusion, we first placed the ultrasound probe over the imaging window and adjusted its position (using micromanipulator) to find back the recording plane from Pre-Stroke session using Bmode (morphological mode) and μ Doppler imaging using brain vascular landmarks (i.e., vascular patterns, brain surface and hippocampus^{34,35} [↗](#), see **Figure 2B** [↗](#)). Functional responses to left whisker stimulation were still detected in the right hemisphere (i.e., contralesional), at the cortical and subcortical levels (orange; **Figure 4A** [↗](#)). As for the early post-stroke imaging period, the functional responses to right whisker stimulation were only detected in the subcortical nuclei and not at the cortical level (green; **Figure 4A** [↗](#)).

Second, we extracted and compared the temporal plots of the functional responses gathered 5d after the stroke with the one obtained from the same two animals at the pre-stroke and 3hrs post-stroke timepoints (**Figure 4B** [↗](#)). At this later time point, the functional responses in the left S1BF (dark green plot, left panel, **Figure 4B** [↗](#). Similar observation were made for the S2 and AuD) remained abolished when compared with the pre-stroke period (black plot), while slightly increased when compared with the 3hrs post-stroke timepoint (green plot). The responses detected in the VPM 5d after the stroke onset (dark green plot, left panel, **Figure 4B** [↗](#)) were largely decreased not only when compared with the pre-stroke signal (black plot) but also with the 3hrs post-stroke trace (green plot). Interestingly, both the amplitude and time-to-peak of the hemodynamic response function were very similar to those from the early post-stroke signal (i.e., 3hrs post-stroke); however, the post-peak period was largely dampened in the 5d post-stroke signal. A similar alteration of the hemodynamic response function was also observed for the 5d post-stroke signal extracted from the Po nucleus when compared to the pre-stroke and 3hrs post-stroke signals (left panel, **Figure 4B** [↗](#). Similar observation were made for the VPL).

Regarding the right hemisphere (i.e., non-ischemic; right panel, **Figure 4B** [↗](#)), the S1BF functional responses to left whisker stimulation were still reduced when compared with pre-stroke responses (black plot) but remained similar to the traces detected at 3-hr post-stroke (orange plot, non-significant). As for the left VPM, both the amplitude and time-to-peak of the hemodynamic responses from the right VPM responses were consistent with pre-stroke and 3hr post-stroke values but the post-peak signal was decreased (brown plot). The functional responses extracted from the Po and VPL did not show changes when compared to pre-stroke and 3hrs post-stroke responses.

Discussion

With this proof-of-concept study, we document on the feasibility of the continuous brain hemodynamics recording of a focal cerebral ischemia after MCAo in conscious rats. Using functional ultrasound imaging, we were able to extract multiple parameters (i.e., ischemia, location and spreading depolarization), characteristic of such cortical stroke. Then, we report on how the functional sensorimotor thalamo-cortical circuit was altered at early and late post-stroke stages.

Compared to highly-invasive conventional strategies such as clipping or suturing^{1,2} [↗](#), the FeCl_3 model used here, is well suited to study stroke under awake conditions. Indeed, the use of FeCl_3 requires less manipulation, allows to maintain the dura intact and strongly reduces the risk of hemorrhage^{36,37} [↗](#) and animal loss. Furthermore, the FeCl_3 model closely mimics key human stroke features including focal ischemia, creation of blood clot, possibility of vessel recanalization, and penumbral tissue^{36,37} [↗](#). However, to adequately and efficiently occlude the vessel of

interest, removing a piece of skull remains required. As mentioned in the report on animal use, one rat was excluded from the analysis as the MCA spontaneously reperfuses, thus dropping the success rate of such model.

The FeCl₃-induced MCAo showed an abrupt and massive drop of blood perfusion remaining constant during the entire recording period. The ischemia was confined within the cortical territory perfused by the MCA (**Figure 2B**), and the infarct (location and size; **Figure 2G**) is in agreement with previous observations^{37,49}. We also detected transient hyperemic events associated with spreading depolarizations (SDs) within the peri-ischemic territory, with occurrence, frequency and amplitude of the hemodynamic waves (**Figure 2D-F**) consistent with prior observation^{43,49,57–61}. Moreover, the spatiotemporal dynamic of the FeCl₃-induced MCAo is consistent with previous fUS imaging reports on cortical ischemia with various stroke model^{16,49,62}.

On top of tracking large hemodynamic variation (i.e. ischemia, SDs), one asset of the fUS imaging technology relies on its ability to track subtle hemodynamic changes in sparse brain regions^{16,26,29,31,33,34,63}. Therefore, we have evaluated how evoked functional responses reorganize at early and late timepoints after stroke induction. Functional responses to mechanical whisker stimulation were detected in several regions relaying the information from the whisker to the cortex, including the VPM and Po nuclei of the thalamus, and S1BF, the somatosensory barrel-field cortex. Responses were also observed in the S2 cortex involved in the multisensory integration of the information^{46,47,64}, the auditory cortex as it receives direct efferent projection from S1BF^{48,64}, and the VPL nuclei of the thalamus connected via corticothalamic projections⁴⁸.

Functional responses extracted in the left hemisphere affected by the focal ischemia (i.e., ipsilesional) show a primary alteration of the whisker-to-barrel pathway within the first hour after the stroke onset. While the abrupt loss in S1BF responses was mainly driven by the focal ischemia, the immediate but partial drop in VPM responses (**Figure 3D**) might result from the direct loss of the excitatory corticothalamic feedback to the VPM^{55,65,66}, or even from a dampening of thalamocortical excitability⁶⁷. The absence of such cortical feedback suggests that the dampened functional responses might be driven by the intrinsic activity of the VPM in response to whisker stimulation. Five days after the initial injury, nuclei of the thalamus (VPM and Po) were subjected to a delayed and robust functional alteration (**Figure 4B**) as previously confirmed in other thalamic relay⁶⁸, probably associated with diaschisis, as previously characterised by tissue staining, reduction of metabolism, functions and perfusion^{21–23,53,68}. Functional responses of the S1BF extracted from the right hemisphere (i.e., contralesional) show a significant decrease shortly after the stroke onset (**Figure 3D**), and still detected at day 5, could be provoked by a loss of transcortical excitability^{69,70}. The late drop in VPM responses might be explained by corticothalamic modulation of the projections toward VP^{46,70}.

While preliminary, these results obtained from awake head-fixed rats are in contradiction with a similar work by our group (fUS imaging, distal MCAo with microvascular clip, electrical whisker stimulation) showing higher contralesional responses to whisker stimulation during early stages of ischemic stroke¹⁶. However, these experiments were subjected to long-term isoflurane regimen (surgery and imaging) known to alter functional responses^{71–73} as well as disrupting hemodynamics⁴⁰. Therefore, further studies will be needed to accurately dissect the complex and long-lasting post-stroke alterations of the functional whisker-to-barrel pathway, including at the neuronal level by direct electrophysiology recordings and imaging, as fUS only reports on hemodynamics as a proxy of local neuronal activity^{30,31,63,74–76}. Another limitation relies on the experimental condition as our brain imaging paradigm was constrained to a single cross-section, thus missing out-of-plane brain regions also affected by the stroke (e.g., ischemic size, infarct extension, origin, and propagation pattern of SDs)⁴¹ or involved in the whisker

network (e.g., superior colliculus, striatum, amygdala and cerebellum)⁴⁶. To overcome such limitation, one can extend the size of the cranial window to allow for larger scale imaging either by sequentially scanning the brain^{30,31,34,35,62,75,77,78}, or by using the recently developed volumetric fUS which provides whole-brain imaging capabilities in anesthetized⁷⁹ and awake rats³³. Finally, it is important to note that this proof-of-concept work did not specifically focus the impact of sex dimorphism on the stroke or early behavioral outcomes following the insult that would greatly enhance the translational value of such preclinical stroke study⁸⁰.

Beyond studying the whisker-to-barrel somatosensory circuit, the brain-wide capability of fUS opens the door to investigate on stroke-affected brain circuits and functions using transgenic lines combined with opto-/chemo-genetic strategies as the technology is fully mature for mice studies^{31,33,34,75}.

Supplementary Materials

Movie 1. Movie of hemodynamic changes induced by MCA occlusion using FeCl₃ in awake head-fixed rats. Raw images.

Movie 2. Movie of thalamo-cortical functional responses to left and right whisker stimulation before and 3hrs after stroke onset.

Author contribution

	Brunner	Montaldo	Urban
Concept	X		
Methodology	X		
Software		X	X
Animal preparation	X		
Imaging	X		
Histochemistry	X		
Data Analysis	X	X	
Visualization	X		
Manuscript	X	X	X
Supervision		X	X
Funding			X

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Animal Tag	Imaging session					Issue
	Control	Baseline (1hr)	Stroke	Early Post-Stroke (3hrs)	Late Post-Stroke (1hr)	
Rat #1	Session excluded	Not performed			Not performed	Miss functional responses from thalamic nuclei as the imaging plane was too anterior
Rat #2	Not performed	Session excluded			Not performed	Unconsistency of functional responses during baseline. Issue with cranial window
Rat #3	Not performed	Session Included			Not performed	
Rat #4	Not performed	Session Included			Not performed	
Rat #5	Session Included	Not performed			Not performed	
Rat #6	Not performed	Session Included			+ 5d Session Included	
Rat #7	Not performed	Session Included			Not performed	
Rat #8	2d before stroke Session Included	Session Included			+ 5d Session Included	
Rat #9	2d before stroke Session Included	Session excluded			Not performed	Reperfusion of the MCA after stroke onset
Sessions used in	- Supplementary Figures 2, 4, 5	- Figures 2 and 3 - Movies 1 and 2 - Supplementary Figures 1 to 5			- Figure 4 - Supplementary Figures 4 and 5	

Supplementary Table 1.

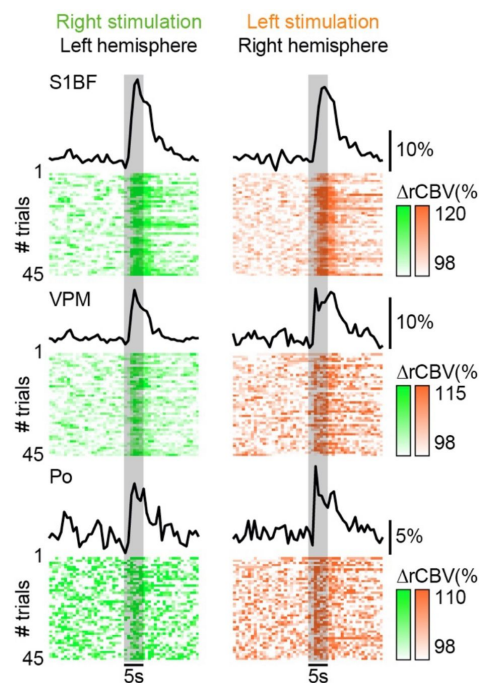
Reporting on animal use, experimentation, exclusion criteria, and figure association.

Region Number	Region Abbreviation	Name of the region
Cortex (Cx)		
1	RSGc	Retrosplenial granular cortex, c region
2	RSD	Retrosplenial dysgranular cortex
3	M2	Secondary motor cortex
4	MPIA	Medial parietal association cortex
5	LPIA	Lateral parietal association cortex
6	S1Tr	Primary somatosensory cortex, trunk region
7	S1DZ	Primary somatosensory cortex, dysgranular zone
8	S1BF	Primary somatosensory cortex, barrel field
9	S2	Secondary somatosensory cortex
10	AuD	Secondary auditory cortex, dorsal area
11	AuI	Primary auditory cortex
12	AuV	Secondary auditory cortex, ventral area
13	Ect	Ectorhinal cortex
14	PRh	Perirhinal cortex
15	DLEnt	Dorsolateral entorhinal cortex
16	DEn	Dorsal endopiriform nucleus
17	Pir	Piriform cortex
18	RAPir	Rostral amygdalopiriform area
19	PLCo	Posterolateral cortical amygdaloid nucleus
20	PMCo	Posteromedial cortical amygdaloid nucleus
Hippocampus (HPC)		
21	DLG	Dorsal lateral geniculate nucleus
22	CA2	Field CA2 of the hippocampus
23	GrDG	Granular layer of the dentate gyrus
24	LMol	Lacunosum moleculare layer of the hippocampus
25	MoDG	Molecular layer of the dentate gyrus
26	PoDG	Polymorph layer of the dentate gyrus
27	FC	Fasciola cinereum
28	Py	Pyramidal cell layer of the hippocampus
29	SLu	Stratum lucidum of the hippocampus
Thalamus (TH) 1/2		
30	LDVL	Laterodorsal thalamic nucleus, ventrolateral part
31	LPMR	Lateral posterior thalamic nucleus, mediorostral part
32	MDL	Mediodorsal thalamic nucleus, lateral part
33	MDC	Mediodorsal thalamic nucleus, central part
34	MDM	Mediodorsal thalamic nucleus, medial part
35	OPC	Oval paracentral thalamic nucleus

Region Number	Region Abbreviation	Name of the region
Thalamus (TH) 2/2		
36	MHb	Medial habenular nucleus
37	LHb	Lateral habenular nucleus
38	PC	Paracentral thalamic nucleus
39	PVP	Paraventricular thalamic nucleus, posterior part
40	IMD	Intermediodorsal thalamic nucleus
41	CL	Centrolateral thalamic nucleus
42	Po	Posterior thalamic nuclear group
43	VPM	Ventral posteromedial thalamic nucleus
44	VPL	Ventral posterolateral thalamic nucleus
45	Rt	Reticular thalamic nucleus
46	VM	Ventromedial thalamic nucleus
47	Rh	Rhomboid thalamic nucleus
48	Re	Reuniens thalamic nucleus
49	Sub	Submedial thalamic nucleus
50	VRe	Ventral reuniens thalamic nucleus
51	ZID	Zona incerta, dorsal part
52	ZIV	Zona incerta, ventral part
53	STh	Subthalamic nucleus
Striatum (STR)		
54	CPu	Caudate putamen
Hypothalamus (HTH)		
55	Subl	Subincertal nucleus
56	A11	A11 dopamine cells
57	A13	A13 dopamine cells
58	PHD	Posterior hypothalamic area, dorsal part
59	DA	Dorsal hypothalamic area
60	DMC	Dorsomedial hypothalamic nucleus, compact part
61	DMD	Dorsomedial hypothalamic nucleus, dorsal part
62	DMV	Dorsomedial hypothalamic nucleus, ventral part
63	PeFLH	Perifornical part of lateral hypothalamus
64	PeF	Perifornical nucleus
65	PLH	Peduncular part of lateral hypothalamus
66	VMHSh	Ventromedial nucleus of the hypothalamus shell
67	ArCL	Arcuate hypothalamic nucleus, lateral part
68	ArCD	Arcuate hypothalamic nucleus, dorsal part
69	ArCM	Arcuate hypothalamic nucleus, medial part

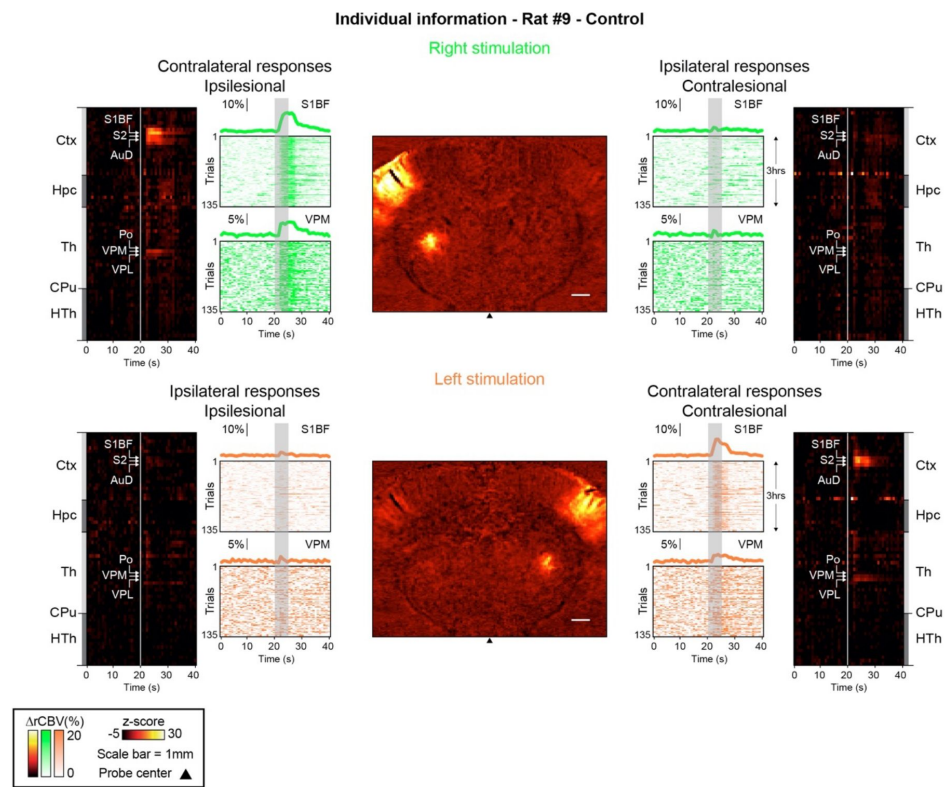
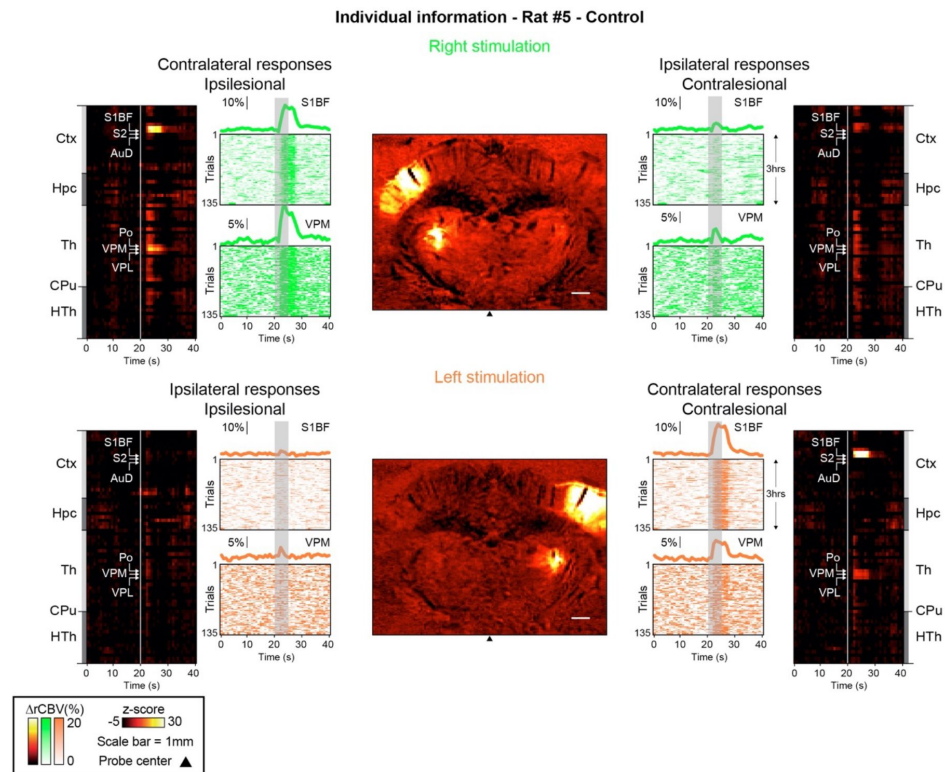
Supplementary Table 2.

List of the 69 brain regions/hemisphere from the coronal cross-section μ Doppler imaged in each rat organized by main anatomical structures. Adapted from the Paxinos rat brain atlas^{42,43}.



Supplementary Figure 2.

Averaged hemodynamic response curves ($\Delta rCBV$ in %) of 45 consecutive right (green) or left whisker stimulation (orange; 1 hr recording) extracted in the contralateral S1BF, VPM and Po regions (top to bottom). The corresponding individual trials presented below confirmed the stability across the recording. Vertical grey bar, the period of whisker stimulation.



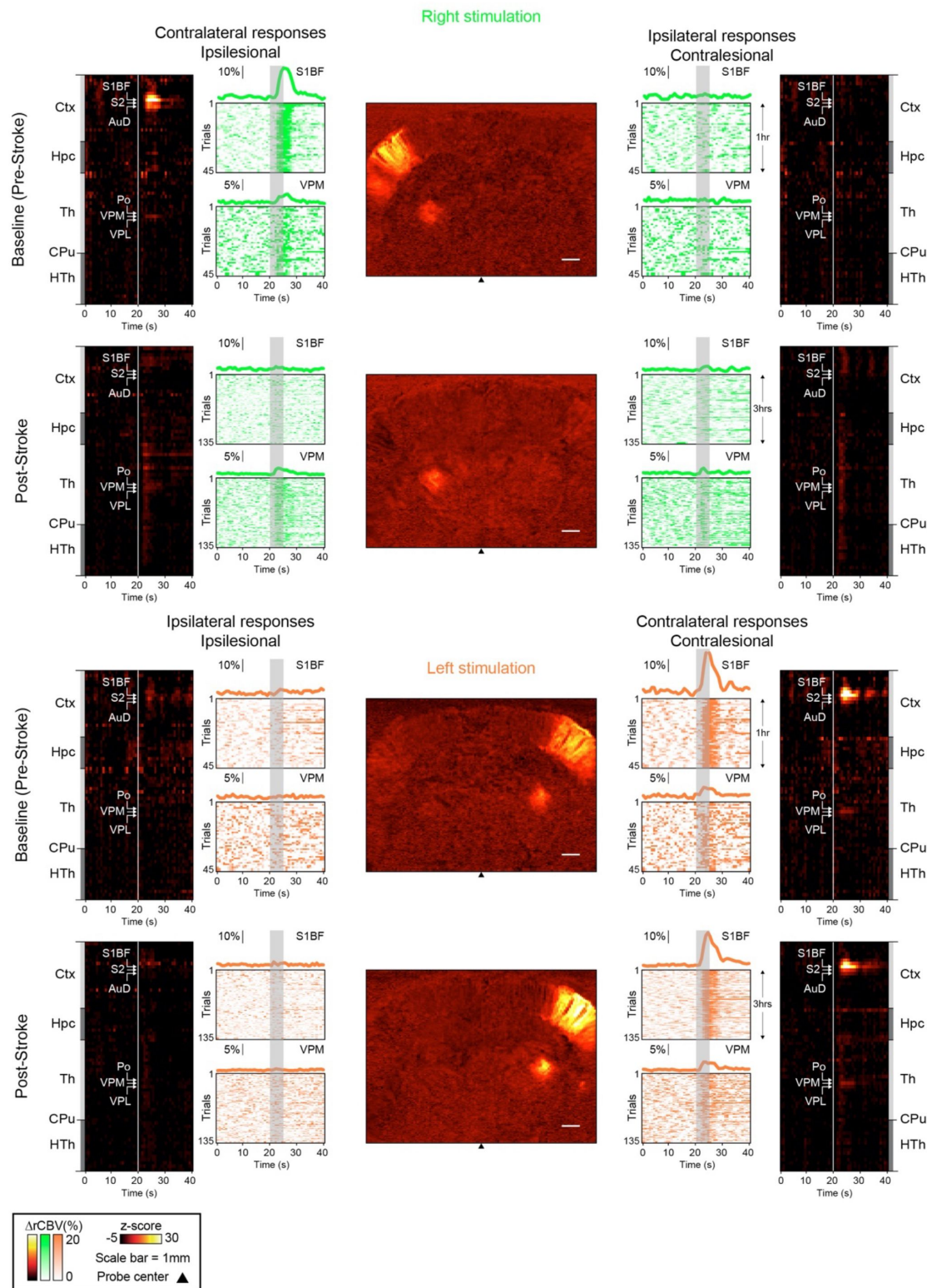
Supplementary Figure 5.

Activity maps, region-time traces of the 69 brain regions imaged, mean and individual time-courses for all trials (left and right stimuli - including contra- and ipsilateral traces) and imaging timepoints (Control, Pre-Stroke, Post-Stroke) for all the rats included in this work.

Right stimulation

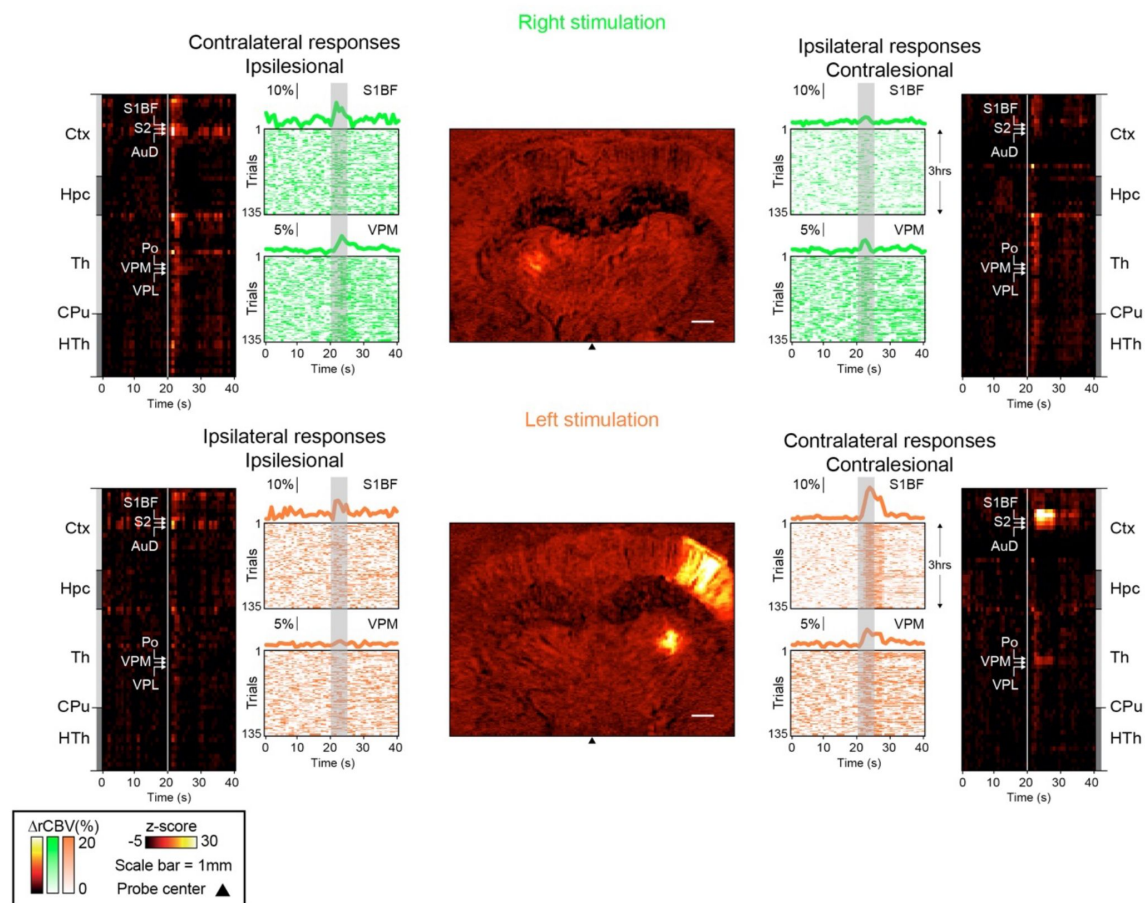


Individual information - Rat #8 - Pre and Post-Stroke



Supplementary Figure 5. (continued)

Individual information - Rat#8 - Post-Stroke 5d



Supplementary Figure 5. (continued)

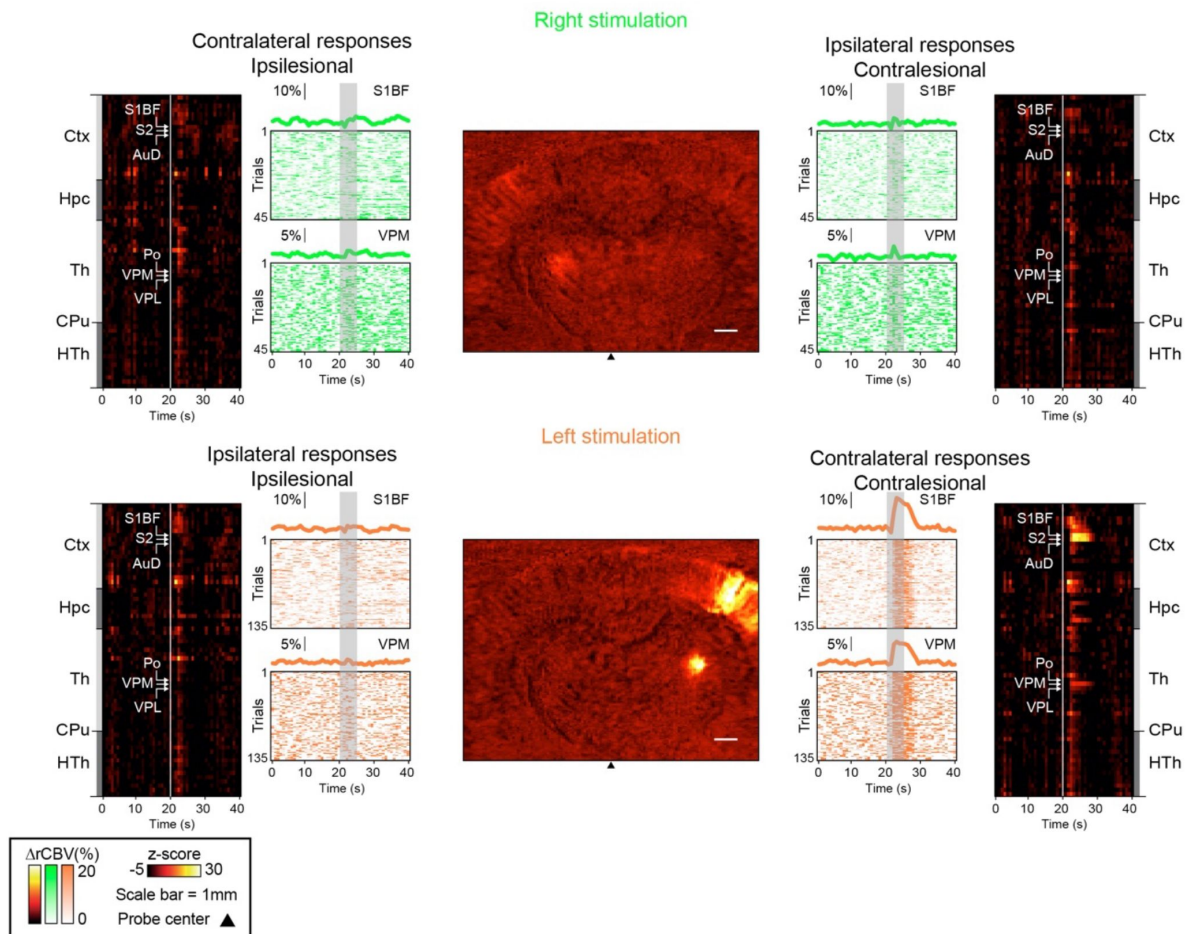
Clément Brunner *et al.*, 2023 eLife. <https://doi.org/10.7554/eLife.88919.2>

Clément Brunner *et al.*, 2023 eLife. <https://doi.org/10.7554/eLife.88919.2>

Right stimulation

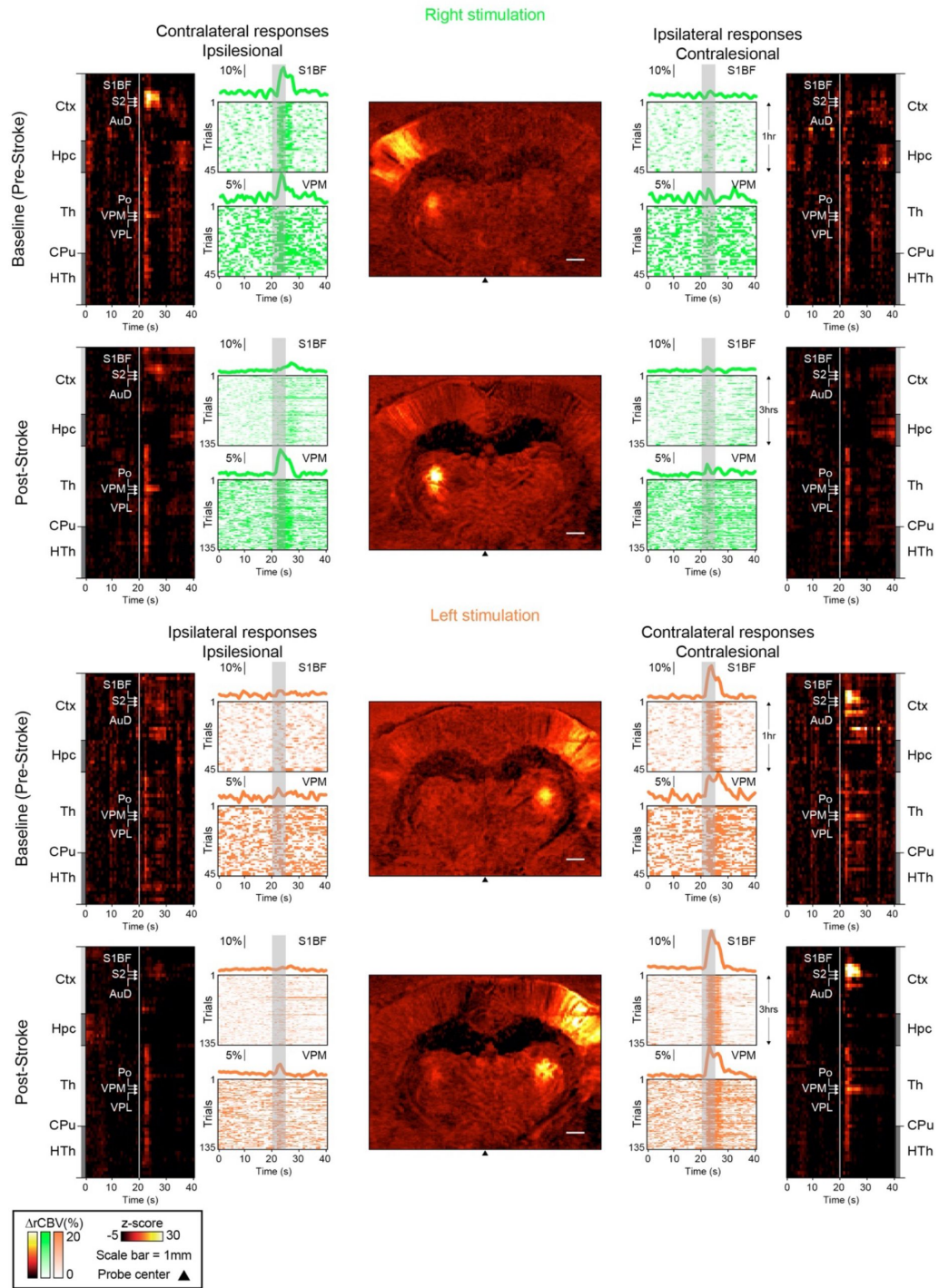


Individual information - Rat #6 - Post-Stroke 5d



Supplementary Figure 5. (continued)

Individual information - Rat #7 - Pre and Post-Stroke



Supplementary Figure 5. (continued)

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Competing interests

A.U. is the founder and a shareholder of AUTC company commercializing functional ultrasound imaging solutions for preclinical and clinical research.

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Editors

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Reviewer #1 (Public Review):

Summary: The authors apply a new approach to monitor widespread changes in sensory evoked hemodynamic activity after focal stroke in fully conscious rats. Using functional ultrasound (fUS), they report immediate and lasting (up to 5 days) depression of sensory evoked responses in somatosensory thalamic and cortical regions.

Strengths: This a technically challenging study that employs new methods to study more distributed changes in sensory evoked neural activity, inferred from changes in cerebral blood flow. The authors provide compelling images and rigorous analysis to support their conclusions.

The primary weakness of this paper was the small sample size used for drawing conclusions. The authors have added additional references that help support their preliminary findings.

Ultimately, it is a proof of concept paper showing the potential of this imaging approach for examining brain wide changes in activity before and after stroke in awake animals. In that sense, I think this paper will be well appreciated by researchers trying to understand how stroke leads to distributed changes in brain function.

Reviewer #2 (Public Review):

Brunner et al. present a new and promising application of functional ultrasound (fUS) imaging to follow the evolution of perfusion and haemodynamics upon thrombotic stroke in awake rats. The authors leveraged a chemically induced occlusion of the rat Medial Cerebral Artery (MCA) with ferric chloride in awake rats, while imaging with fUS cerebral perfusion with high spatial and temporal resolution ($100\mu\text{m} \times 110\mu\text{m} \times 300\mu\text{m} \times 0.8\text{s}$). The authors also measured evoked haemodynamic responses at different timepoints following whisker stimulation.

As the fUS setup of the authors is limited to 2D imaging, Brunner and colleagues focused on a single coronal slice where they identified the primary Somatosensory Barrel Field of the Cortex (S1BF), directly perfused by the MCA and relay nuclei of the Thalamus: the Posterior (Po) and the Ventroposterior Medial (VPM) nuclei of the Thalamus. All these regions are involved in the sensory processing of whisker stimulation. By investigating these regions the authors present the hyper-acute effect of the stroke with these main results:

- MCA occlusion results in a fast and important loss of perfusion in the ipsilesional cortex.
- Thrombolysis is followed by Spreading Depolarisation measured in the Retrosplenial cortex.
- Stroke-induced hypo-perfusion is associated with a significant drop in ipsilesional cortical response to whisker stimulation, and a milder one in ipsilesional subcortical relays.
- Contralesional hemisphere is almost not affected by stroke with the exception of the cortex which presents a mildly reduced response to the stimulation.

In addition, the authors demonstrate that their protocol allows to follow up stroke evolution up to five days postinduction. They further show that fUS can estimate the size of the infarcted volume with brilliance mode (Bmode), confirming the presence of the identified lesional tissue with post-mortem cresyl violet staining.

Upon measuring functional response to whisker stimulation 5 days after stroke induction, the authors report that:

- The ipsilesional cortex presents no response to the stimulation
- The ipsilesional thalamic relays are less activated than hyper acutely

These observations mainly validate a new method to chronically image the longitudinal sequelae of stroke in awake animals. However, the potentially more intriguing results the authors describe in terms of functional reorganization of functional activity following stroke will require additional data to be validated. While highly preliminary, the research model proposed by the author (where the loss of the infarcted cortex induces reduces activity in connected regions, whether by cortico-thalamic or cortico-cortical loss of excitatory drive), is interesting. This hypothesis would require a greatly expanded, sufficiently powered study to be validated (or disproven)."

Reviewer #3 (Public Review):

The authors set out to demonstrate the utility of functional ultrasound for evaluating changes in brain hemodynamics elicited acutely and subacutely by middle cerebral artery occlusion model of ischemic stroke in awake rats.

Functional ultrasound affords a distinct set of tradeoffs relative to competing imaging modalities. Acclimatization of rats for awake imaging has proven difficult with most, and the

high quality of presented data in awake rats is a major achievement. The major weakness of the approach is in its being restricted to single slice acquisitions, which also complicates registration of acquisition across multiple imaging sessions within the same animal. Establishing that awake imaging represents an advancement in relation to studies under anesthesia hinges upon establishment of the level of stress experienced by the animals in the course of imaging, i.e., requires providing data on the assessment of stress over the course of these long imaging sessions, which was not undertaken. This is particularly significant given that physical restraint has been established to be a particularly potent stressor in experimental models of stress. Assessment of the robustness of these measurements in a larger cohort of animals under varying conditions is of particular significance for supporting its wide applicability.

Author Response

The following is the authors' response to the original reviews.

Thank you for your consideration and insightful comments on our article.

We have gone through all the reviewers' comments and addressed all their questions and concerns point by point.

As per their recommendation, we have amended our manuscript by providing more information about the experimental procedure and statistical analysis followed, and removed some analyses with a reduced number of imaging sessions. In addition, as a Resource and Tools article, the claim of our paper has been adjusted to a proof-of-concept paper showing robust and reliable preliminary results. In the meantime, we have provided 3 new Supplementary Figures, including one showing data from all individual animals.

Reviewer #1 (Public Review):

The authors apply a new approach to monitor brain-wide changes in sensory-evoked hemodynamic activity after focal stroke in fully conscious rats. Using functional ultrasound (fUS), they report immediate and lasting (up to 5 days) depression of sensory-evoked responses in somatosensory thalamic and cortical regions.

Strengths: This a technically challenging and proof-of-concept study that employs new methods to study brain-wide changes in sensory-evoked neural activity, inferred from changes in cerebral blood flow. Despite the minor typos/grammatical errors and small sample size, the authors provide compelling images and rigorous analysis to support their conclusions. Overall, this was a very technically difficult study that was well executed. I believe that it will pave the way for more extensive studies using this methodological approach. Therefore I support this study and my recommendations to improve it are relatively minor in nature and should be simple for the authors to address.

Weaknesses: The primary weakness of this paper is the small sample sizes. Drawing conclusions based on the small sham control group (n=2) or 5-day stroke recovery group (n=2), is rather tenuous. One way to alleviate some uncertainty with regard to the conclusions would be to state in the discussion that the findings (ie. loss of thalamocortical function after stroke) are perfectly consistent with previous studies that examined thalamocortical function after stroke. The authors missed some of these supporting studies in their reference list (see PMID: 28643802, 1400649). A second issue that can easily be resolved is their analysis of the 69 brain regions. This seems like a very important part of the study and one of the primary advantages of employing efUS. As presented, I had difficulty seeing the data. I think it would be worthwhile to expand Fig 3 (especially 3C) into a full-page figure with an accompanying table in the Supplementary info section describing the % change in CBF for each brain region.

Other Recommendations for the authors:.

- *Since there is variability in spreading depolarizations, was there any trend in the relationship between # SD's and ischemic volume? I know there are few data points but a scatterplot might be of interest.*
- *For statistical comparisons of 'response curves' in Fig 3 and 4, what exactly was the primary dependent measure: changes in peak amplitude (%) or area under the curve?*
- *There are several typos and minor grammatical errors in the manuscript. Some editing is recommended.*

We thank the reviewer for the comments and suggestion, we have adapted our message to a proof-of-concept paper showing robust and reliable preliminary results. We also thank the reviewer for pointing out important references that support our observation and have added them to our article. We have provided a supplementary full-page version of the current Figure 3C (see Supplementary Figure 3).

Regarding the recommendations, we strongly agree that it would be of interest to link SDs and ischaemia, but unfortunately this can't be done because our experimental design, i.e. narrow cranial window and single static plane, does not allow brain-wide quantification of ischemic volume. This would be possible either by scanning the brain or by using a matrix array (also discussed in the manuscript).

For statistical analysis of the hemodynamic response curves, we have adapted them to compare the area under the curve (AUC). In addition, we have provided a new Supplementary Figure 4 showing the associated values and statistics.

We have edited typos and errors.

Reviewer #2 (Public Review):

Brunner et al. present a new and promising application of functional ultrasound (fUS) imaging to follow the evolution of perfusion and haemodynamics upon thrombotic stroke in awake rats. The authors leveraged a chemically induced occlusion of the rat Medial Cerebral Artery (MCA) with ferric chloride in awake rats, while imaging with fUS cerebral perfusion with high spatio and temporal resolution (100µm x 110µm x 300µm x 0.8s). The authors also measured evoked haemodynamic response at different timepoints following whisker stimulation.

As the fUS setup of the authors is limited to 2D imaging, Brunner and colleagues focused on a single coronal slice where they identified the primary Somatosensory Barrel Field of the Cortex (S1BF), directly perfused by the MCA and relay nuclei of the Thalamus: the Posterior (Po) and the Ventroposterior Medial (VPM) nuclei of the Thalamus. All these regions are involved in the sensory processing of whisker stimulation. By investigating these regions the authors present the hyper-acute effect of the stroke with these main results:

- *MCA occlusion results in a fast and important loss of perfusion in the ipsilesional cortex.*
- *Thrombolysis is followed by Spreading Depolarisation measured in the Retrosplenial cortex.*

- Stroke-induced hypo-perfusion is associated with a significant drop in ipsilesional cortical response to whisker stimulation, and a milder one in ipsilesional subcortical relays.
- Contralesional hemisphere is almost not affected by stroke with the exception of the cortex which presents a mildly reduced response to the stimulation.

In addition, the authors demonstrate that their protocol allows to follow up stroke evolution up to five days post-induction. They further show that fUS can estimate the size of the infarcted volume with brilliance mode (B-mode), confirming the presence of the identified lesional tissue with post-mortem cresyl violet staining.

Upon measuring functional response to whisker stimulation 5 days after stroke induction, the authors report that:

- *The ipsilesional cortex presents no response to the stimulation*
- *The ipsilesional thalamic relays are less activated than hyper acutely*
- *The contralesional cortex and subcortical regions are also less activated 5d after the stroke.*

These observations mainly validate the new method as a way to chronically image the longitudinal sequelae of stroke in awake animals. However, the potentially more intriguing results the authors describe in terms of functional reorganization of functional activity following stroke appear to be preliminary, and underpowered (N = 5 animals were imaged to describe hyper-acute session, and N = 2 in a five day follow-up). While highly preliminary, the research model proposed by the author (where the loss of the infarcted cortex induces reduces activity in connected regions, whether by cortico-thalamic or cortico-cortical loss of excitatory drive), is interesting. This hypothesis would require a greatly expanded, sufficiently powered study to be validated (or disproven).

We thank the reviewer for the careful and accurate description of our work. We have addressed all the comments, recommendations and concerns raised by providing details of the experimental procedure and statistical analysis followed, and by removing some analyses associated with a reduced number of imaging sessions (at d5, n=2).

Reviewer #3 (Public Review):

The authors set out to demonstrate the utility of functional ultrasound for evaluating changes in brain hemodynamics elicited acutely and subacutely by the middle cerebral artery occlusion model of ischemic stroke in awake rats.

Functional ultrasound affords a distinct set of tradeoffs relative to competing imaging modalities. Acclimatization of rats for awake imaging has proven difficult with most, and the high quality of presented data in awake rats is a major achievement. The major weakness of the approach is in its being restricted to single-slice acquisitions, which also complicates the registration of acquisition across multiple imaging sessions within the same animal. Establishing that awake imaging represents an advancement in relation to studies under anesthesia hinges upon the establishment of the level of stress experienced by the animals in the course of imaging, i.e., requires providing data on the assessment of stress over the course of these long imaging sessions. This is particularly significant given how significant a stressor physical restraint has been established to be in rodent

models of stress. Furthermore, assessment of the robustness of these measurements is of particular significance for supporting the wide applicability of this approach to preclinical studies of brain injury: the individual animal data (effect sizes, activation areas, kinetics) should thus be displayed and the statistical analysis expanded. Both within-subject, within/across sessions, and across-subjects variability should be evaluated. Thoughtful comments on the relationship between power doppler signal and cerebral blood volume are important to include and facilitate comparisons to studies recording other blood volume-weighted signals. Finally, the contextualization of the observations with respect to other studies examining acute and subacute changes in brain hemodynamics post focal ischemic stroke in rats is needed. It is also quite helpful, for establishing the robustness of the approach, when the statistical parametric maps are shown in full (i.e. unmasked).

We would like to thank the reviewer for the comments, recommendations and concerns he/she/they raised. We have addressed all the points to clarify our article and make it more relevant and informative for readers.

Reviewer #2 (Recommendations For The Authors):

The work described by Brunner et al is primarily a methodological paper, with potentially interesting, yet not robust enough, novel biological insight into the mechanisms of stroke. Nonetheless, the method employed is interesting and potentially well-validated.

General comments/suggestions

1- One potential concern I have is related to the relatively low sample size used, with $n=5$ for the main results and only $n=2$ for the follow-up after 5d. I am not sure much can be generalized using only two animals in any research study and this $N = 2$ dataset should probably be removed entirely from the study. Moreover, I found the statistical methods used were only superficially described, which prevented me from assessing whether the results reported by the authors are biologically relevant or not (including some significant differences in rCBV well below 1% estimated over two individuals).

We fully agree with the reviewer's comment and balanced our claim by considering this work as a proof-of-concept on brain imaging of multiple aspects of stroke hemodynamics (ischemia, spreading depolarization-like events, cortico-thalamic functions) in awake head-fixed rats. Therefore, we attenuated our message along the entire manuscript to prevent misunderstanding and over statement (e.g., Lines 356, 441, 455), we also remove statistics from the analysis at d5 post-stroke, see Figure 4 and associated paragraph from Line 356.

2- Based on their investigations, the authors propose a model where the loss of infarcted cortex induces reduced activity in connected regions, whether by cortico-thalamic or cortico-cortical loss of excitatory drive. This is an intriguing framework but this hypothesis would require a more complete, well-powered study to be substantiated.

I think a clear recognition of the fact that these findings are just preliminary and not validated should be more explicitly reported. I also marginally note here that these results are in contrast with previous reports from the same team where occlusion of the MCA induced increased response to whisker stimulation in anaesthetised rats. These contradictory findings are not discussed in this manuscript.

As mentioned above, we explicit more on the proof-of-concept proposed in this work as well as clearly stating on the preliminary aspect of the findings described in this work. As mentioned above, we attenuated our message along the entire manuscript to prevent

misunderstanding and over statement (e.g., Lines 348, 433, 447), we also remove statistics from the analysis at d5 post-stroke, see figure 4 and associated paragraph from Line 348.

We thanks the reviewer for pointing out the missing link with our previous work performed under anesthesia. We therefore provided a discussion point on this contradictory finding (Line 441).

3- In a previous study from the same group perfusion was imaged in 3D either by means of a motorized probe or by using a 2D matrix arrays. It would be interesting to discuss why a 2D approach was chosen in this study over those previous methods.

Indeed, brain-wide coverage would be of great interest in such experiment context. As mentionned by the reviewer, two strategies can be used:

- One can scan the brain using a motorized probe as performed for different purposes by Sieu et al., Nature Methods, 2015; Hingot, Brodin et al., Theranostics 2020; Macé et al., Neuron 2019 and also by our group in Sans-Dublanç, Chrzanowska et al., Neuron, 2022; Brunner et al. Frontiers in Neuroscience 2022 and Brunner et al., JCBFM 2023. (This list of publication is not exhaustive).
- A second approach aims at using a 2D matrix array to capture functions at brain-wide scale. So far, this strategy has been employed in a couple of studies (Rabut et al., Nature Methods, 2019 and Brunner, Grillet et al., Neuron, 2020).

The strategy consisting of scanning (manually or using a motor) strongly limits investigation on brain functions, as performing an accurate covering of the functional regions requires an extensive and time-consuming scanning: brain functions must be addressed several time to capture a reliable and robust signal for all the brain section scanned (see Brunner et al., 2022). Unfortunately, this strategy prevents us to accurately capture other brain hemodynamics like the dynamic of the ischemia or the spreading depolarization event.

On the other hand, the volumetric functional ultrasound imaging (vfUSI) would be suited for brain-wide coverage capturing large-scale brain functions (see Brunner, Grillet et al. Neuron 2020) and hemodynamic events (see Rabut et al., Nature Methods, 2019) but at the cost of the resolution, frame rate and larger cranial window. Unfortunately, this technology was not available when this work was conducted.

Such experimental opportunities have been suggested at the end of the manuscript: “To overcome such limitation, one can extend the size of the cranial window to allow for larger scale imaging either by sequentially scanning the brain^{27,28,31,32,59,69,71,72}, or by using the recently developed volumetric fUS which provides whole-brain imaging capabilities in anesthetized⁷³ and awake rats³⁰.”

4- Overall the registration scheme seems suboptimal which ultimately questions the specificity of the findings in thalamic regions. It would be interesting to validate this procedure, especially the probe repositioning five days after the stroke.

Positioning was not difficult part of this experiment. First, all head posts were implanted in the same position relative to the skull references bregma and lambda. Second, the head fixation ensures the same placement of the headpost for all animals. Finally, fine adjustment of the ultrasound probe position were done using a micromanipulator by finding key landmarks from the μ Doppler image. In practice, minimal adjustments were needed to find back the same imaging plane. We provide additional information about the positioning in the Materials and Methods section.

New text – Line 126: “Positionning.

The mechanical fixation of the head-post ensures an easy and repeatable positioning of the ultrasound probe across imaging session. The ultrasound probe is indeed fixed to a micromanipulator enabling light adjustments To find the plane of interest (containing both S1BF and thalamic relays: bregma - 3.4mm), we used brain landmarks (e.g., surface of the brain, hippocampus, superior sagittal sinus, large vessels). Note that as the headpost was carefully placed in the same position relative to the skulls landmarks (bregma and lambda), the position of the region of interest was minimal across animals.”

Second, at d5 post-stroke, we positionned the ultrasound probe over the imaging window as described in the Materials and Methods section and use brain landmarks from baseline/post-stroke image to maximize the position of brain image. We better detail the procedure followed.

Original text: “First, we used the vascular markers and the shape of the hippocampus^{31,32} to find back the coronal cross-section imaged during the pre-stroke session. Five days after the MCA occlusion,...”

New text – Line 360 :“Five days after the MCA occlusion, we first placed the ultrasound probe over the imaging window and adjusted its position (using micromanipulator) to find back the recording plane from Pre-Stroke session using Bmode (morphological mode) and μ Doppler imaging using brain vascular landmarks (i.e., vascular patterns, brain surface and hippocampus^{34,35}; see Figure 2B).”

More detailed questions/comments/suggestions

Methods

ARRIVE methodology

- *Point 2b: sample size is not adequately explained, especially the use of $n = 2$ animals for 5d follow up*

We have explicated the sample size by adding a short paragraph at the beginning of the Results section. We also make the Supplementary Table 1 more accurate. New text – Line 239: “Animals

Report on animal use, experimentation, exclusion criteria can be found in Supplementary Table 1. Rat#1 was excluded after the control session as the imaging window was too anterior to capture both cortical and thalamic responses. Ra#2 was excluded as hemodynamic responses were inconsistent during baseline (pre-stroke) period. Rat#3 showed early post-stroke reperfusion and was excluded from stroke analysis, the control session (pre-stroke) from Rat#3 was analyzed.”

- *Point 7: statistical methods: The quantification used to assess significant differences in stimulation traces is poorly described.*

We have amended the Materials and Methods section about statistics and provided Supplementary Figure 4.

New text – Line 221: “Activated brain regions were detected from hemodynamic response time-courses using GLM followed by t-test across animals as proposed in Brunner, Grillet et al.,³⁴. The area under the curve (AUC) from hemodynamic response time-courses was computed for individual trials in S1BF, VPM and Po regions, for all the periods of the recording and for all rats included in this work. AUC were compared and analysed using a

non-parametric Kruskal-Wallis test corrected for multiple comparison using a Dunn's test. Tests were performed using GraphPad Prism 10.0.1. “

Functional Ultrasound Imaging acquisition

- References 26 and 28 imply 2.5Hz and 2Hz acquisition rates, respectively. Why does the same method result in a 1.25Hz acquisition rate here? Can you confirm the same spatial resolution in these conditions?

The spatial resolution is independent of the temporal resolution (frame rate). The spatial resolution depends on the resolution of the compound image and the temporal resolution is given by the number of compound images to generate a single Doppler image (exposure time). By increasing the number of compound images, the frame rate decreases while increasing the signal to noise ratio and sensitivity. For some work, a pause between 2 frames is used (mostly due to technical limitations in the software (processing time , or execution of a real-time display/processing by the user), however this reduces the frame rate.

Author response table 1.

Comparing with the sequences used in references 26 and 28, we have the following timing parameters

Sequence	Frequency (Hz)	N Compound	Acquisition time(s)	Pause (s)	Frame rate(Hz)
Maci et al.	High sensibility	500	250	0.4	0.2
Neumaier et al.	High sensibility	500	250	0.1	0.1
Brunner et al.	High sensibility	500	250	0.5	0
Neumaier et al.	High sensibility	500	250	0.1	0
This study	High sensibility	500	250	0.5	0.3

In this work, we decided to reduce the frame rate to have less images but with higher SNR. The 0.3s were added by technical considerations in this specific implementation.

New text – Line 158:“ To obtain a single vascular image we acquired a set of 250 compound images in 0.5s, an extra 0.3s pause is included between each image to have some processing time to display the images for real-time monitoring of the experiment. “

Activity Maps

- How is the use of a 40s window motivated?

The 40s window has been chosen to better compare hemodynamic responses to either left or right whisker stimulation and centered the period of interest on the start of the stimulation. Original text:” Pre- and post-stroke recordings are reshaped in shorter 40-s sessions, i.e., 50 frames, ...”

New text – Line 206:“ Pre- and post-stroke recordings are reshaped in 40-s sessions, i.e., 50 frames, centered on the start of the stimulation (at 20s), ...”

- I think the manuscript would benefit from the use of an established, event-based GLM for activity mapping.

We thank the reviewer for this suggestion, here we used a z-score for activity mapping that is largely established in the neuroimaging realm.

- The statistical thresholds used should account for multiple comparisons.

We have amended the Materials and Methods section, and figure captions about statistics and provided Supplementary Figure 4.

Statistical analyses

- *Overall this section is only superficially described, and lacks detailed information.*

We have amended the Materials and Methods section about statistics and provided Supplementary Figure 4.

New text – Line 221 : “Activated brain regions were detected from hemodynamic response time-courses using GLM followed by t-test across animals as proposed in Brunner, Grillet et al.,³⁴. The area under the curve (AUC) from hemodynamic response time-courses was computed for individual trials in S1BF, VPM and Po regions, for all the periods of the recording and for all rats included in this work. AUC were compared and analysed using a non-parametric Kruskal-Wallis test corrected for multiple comparison using a Dunn’s test. Tests were performed using GraphPad Prism 10.0.1. “

- *Are average rCBV changes referred to in the 40s window?*

The rCBV changes are referring to the pre-stimulation baseline. We have modified the text accordingly (Line 206).

- *Were normality and variance equality requirements verified in the group with n=2?*

Based on reviewers comment’s on the limited amount of recording at 5d, we have decided to remove this statistical analysis. The manuscript, figure and caption were corrected accordingly.

- *There is no method for cresyl violet staining*

We thank the review for highlighting this omission. We have provided a paragraph in the Materials & Methods section detailing the histology procedure – Line 228:

“Histopathology

Rats were killed 24hrs after the occlusion for histological analysis of the infarcted tissue. Rats received a lethal injection of pentobarbital (100mg/kg i.p. Doletal, Vetoquinol, France). Using a peristaltic pump, they were transcardially perfused with phosphate-buffered saline followed by 4% paraformaldehyde (Sigma-Aldrich, USA). Brains were collected and post-fixed overnight. 50-µm thick coronal brain sections across the MCA territory were sliced on a vibratome (VT1000S, Leica Microsystems, Germany) and analyzed using the cresyl violet (Electron Microscopy Sciences, USA) staining procedure (see Open Lab Book for procedure). Slices were mounted with DPX mounting medium (Sigma-Aldrich, USA) and scanned using a bright-field microscope.”

Results 1: Real time imaging of stroke induction in awake rats

- *Why is the window so narrow in the anteroposterior direction?*

The imaging window was defined based on the brain regions investigated in this work, meaning the primary somatosensory cortex (S1BF) and the ventroposterior medial thalamic

relay (VPM). From Paxinos atlas, a position of interest is located at Bregma -3.4mm. The cranial window was performed accordingly, and restricted couple of mm to avoid non-needed procedure and brain exposure. We added a new sentence in the Materials & Methods section – Line 116: “This cranial window aims to cover bilateral thalamo-cortical circuits of the somatosensory whisker-to-barrel pathway.”

- *What validation was employed for the habituation protocol? Are animals stressed by the procedure? Do you have cortisol data to show? Are animal weights throughout the procedure?*

The habituation protocol employed in this work follows recommendations from the expert in the field and peers (Martin et al., Journal of Neuroscience Methods, 2002; Martin et al., Neuroimage 2006; Topchiy et al., Behav Brain Res 2009). We have amended the corresponding paragraph in the Materials & Methods section detailing the habituation procedure:

Original text: “Body restraint and head fixation.

Rats were habituated to the workbench and to be restrained in a sling suit (Lomir Biomedical inc, Canada), progressively increasing the restraining period from minutes to hours^{33,34}. After the headpost implantation (see below), rats were habituated to be head-fixed while restrained in the sling. The period of fixation was progressively increased from minutes to hours. Water and food gel (DietGel, ClearH2O, USA) were provided along the habituation session. Once habituated, the cranial window for imaging was performed as described below (Figure 1A-C).”

New text - Line 90: “Body restraint and head fixation.

The body restraint and head fixation procedures are adapted from published protocols and setup dedicated for brain imaging of awake rats^{39–41}. Rats were habituated to the workbench and to be restrained in a sling suit (Lomir Biomedical inc, Canada) by progressively increasing restraining periods from minutes (5mins, 10mins, 30mins) to hours (1 and 3hrs) for one or two weeks. The habituation to head-fixation started by short (5 to 30s) and gentle head-fixation of the headpost between fingers. The headpost was then secured between clamps for fixation periods progressively increased following the same procedure as with the sling. For both body restraint and head fixation, the initial struggling and vocalization diminished over sessions. Water and food gel (DietGel, ClearH2O, USA) were provided for all body restraint and head-fixation habituation sessions. Once habituated, the cranial window for imaging was performed as described below (Figure 1A-C).”

- *The observation of contralateral oligemia is based only on RSG traces.*

We provided contralesional perfusion changes for all regions in Supplementary Figure 1.

- *The spatial and temporal distribution of Bmode measured hyperechogenicity is surprising and should be discussed. Reference 29 describes for instance non-overlap with an area of hypo-perfusion. Overlap between hypo-perfused and infarct volumes should be systematically investigated and coregistered with histology. Moreover, reference 40, while using a different model, presents hyperechogenicity at 5h.*

The B-mode images in Figure 2B are presented as an illustration of the potential morphological changes detected at different timepoint. However, our study focuses on functional responses and not on the evolution of the morphological changes. Indeed, this Bmode images remain difficult to interpret as they show a structural reorganization at the

level of the ultrasound scatterers which has not been directly linked with tissue infarction, oedema, or other histological conditions.

Regarding the reference 40, the authors found an hyper-echogenicity at 5h a time window is not covered by our protocol. In reference 29, we indeed detailed a mismatch between the μ Doppler images and histopathology. As suggested by the reviewer, seeking for other potential mismatches/overlaps between Bmode/ μ Doppler and histopathology is an interesting field on investigation, but remains out of the scope of this work.

Results 3: Delayed alteration of the somatosensory thalamocortical pathway

- *These results are underpowered and as such should probably be removed entirely from the paper (or substantiated with greater Ns of animals). Based on reviewers comment's on the limited amount of recording at 5d, we have decided to remove this statistical analysis. The manuscript, figure and caption were corrected accordingly.*
- *If I am not mistaken, reference 28 describes a protocol for awake mouse imaging, and thereby does not introduce any hippocampal landmark allowing effective positioning of the probe.*

We thanks the reviewer for this comment. While not used in the figure detailing image registration in reference 28, step 42 (page 17) from the protocol mentions the use of hippocampal landmark to position of the imaged brain to the atlas. The hippocampal landmark is also used in Brunner et al., JCBFM 2023, we have added this reference which is more appropriate to this work (i.e., rat model, digitalized paxinos atlas, linear ultrasound transducer).

- *Significant difference in ipsilesional VPM with post-stroke period looks spurious.*

We have amended the Materials and Methods section about statistics and provided Supplementary Figure 4.

Discussion:

The sentence "might result from the direct loss of the excitatory corticothalamic feedback to the VPM" should be moderated in the absence of electrophysiology support. Such a decrease could be explained by reduced perfusion due to the challenge.

The reviewer is right and we believe the tense used in the sentence already balance the claim. However, we clarified on how such result could be better validated.

Original text: "Further work will need to dissect the complex and long-lasting post-stroke alterations of the functional whisker-to-barrel pathway, including at the neuronal level, as fUS only reports on hemodynamics as a proxy of local neuronal activity^{27,28,60,66–68}"

New text – Line 445: "Therefore, further studies will be needed to accurately dissect the complex and long-lasting post-stroke alterations of the functional whisker-to-barrel pathway, including at the neuronal level by direct electrophysiology recordings and imaging, as fUS only reports on hemodynamics as a proxy of local neuronal activity^{30,31,63,74–76}."

Figure 2

- *Panel B would be more informative if presented as an average.*

The aim of this figure is to show the raw data of a typical case. Averaging μ Doppler images wouldn't be illustrative as individual vessels will not be visible anymore. Because the vessels are in different positions from one animal to another, an average image would be blurred.

- *Panel C lacks contralateral S1BF trace.*

We have provided contralesional perfusion changes for all regions in Supplementary Figure 1.

- *Methods for detection of SDs refer to non-peer-reviewed reference 29, where SD is defined as 50% over baseline level. What is the actual threshold/method used to define a SD in this study?*

We better detailed this procedure in the Materials & Methods section - Line 195: "The detection of hemodynamic events associated with spreading depolarizations (SDs) was performed based on the temporal analysis of the rCBV signal in the retrosplenial granular (RSGc) and dysgranular (RSD) cortices of the left hemisphere (ipsi-lesional). SDs were defined as transient increase of rCBV signal (+25%) detected with a temporal delay of <10 frames (i.e., 8secs) between the two regions of interest, validating both the hyperemia and spreading features of hemodynamic events associated with spreading depolarizations."

- *For panel F, a measure of variance would be more suited to show stereotypic profile across animals as the number of SDs varies between animals.*

Figure 2F indeed shows the average profile of hemodynamic events associated with spreading depolarizations (black line) with the variance (95% confidence interval error bands in gray). We have adjusted the corresponding figure caption to make this information more clear.

Figure 3

- *The exact stimulation employed is not clear as the methods describe a 1.33 min delay between two whisker pad stimulations, but the figure reports 40s. The description is thereby ambiguous. We thank the reviewer for pointing out this potential confusion which allowed us to correct a mistake*

- The effective delay between two stimulations delivered to the whisker pads is 40 seconds
- The effective delay between two stimulations delivered to the same whisker pad is 80 seconds from start to start or 75 seconds from end to start.

The text was amended accordingly in line 144: "Thus, the effective delay between two stimulations delivered to the same whisker pad is 80 seconds from start to start."

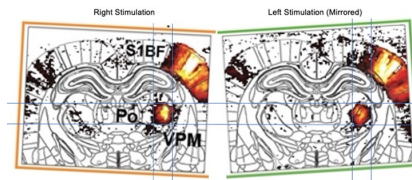
- *In panel B the choice of colormap and transparency for template overlay is not explained and is confusing given the employed threshold of 1.6. Which mask was used to overlay the activation map on the template? Why black color to represent a supposedly significant difference?*

We thank the reviewer for pointing out this potential confusion. We have adjusted the colormap in Figures 3 and 4.

- *The pre-stroke thalamic response is clearly localized in VPM for left stimulation, while it overlaps VPM and Po for the right stimulation. This questions the accuracy of the employed registration scheme and consequently the choice of these ROIs, which appear quite small as compared to the resolution and this positioning precision.*

We see the point of the reviewer, here the apparent difference because the brain is slightly tilted. By adjusting the angle for both activity maps (see Author response image 1) we confirm that both maps are very similar including the for activated areas VPM and Po.

Author response image 1.



- *It would be interesting to see the same activation maps for all animals in supplementary.*

We have provided the Supplementary Figure 5 that contains both ipsilateral and contralateral responses to whiskers stimulation (from both left and right pads) for all trials and all rats included in this work.

- *Looking at panel C, more cortical regions seem to respond to the stimulation above S1BF.*

The reviewer is right and we have indeed mentioned this point several times in the original manuscript in:

- the result section: “We also detected significant increase of activity in S2, AuD, Ect (**** $p < 0.0001$) and PRh (* $p < 0.001$) **cortices and VPL nucleus** ($p < 0.01$; the list of acronyms is provided in Supplementary Table 2), brain regions receiving direct efferent projections from the S1BF45,48,49, VPM or Po nuclei50–52.”
- the caption of Figure 4: “S1BF, S2, AuD, VPM, VPL and Po regions are brain regions significantly activated (all p value <0.01 ; GLM followed by t-test.”
- the conclusion section : “Functional responses to mechanical whisker stimulation were detected in several regions relaying the information from the whisker to the cortex, including the VPM and Po nuclei of the thalamus, and S1BF, the somatosensory barrel-field cortex. Responses were also observed in the S2 cortex involved in the multisensory integration of the information43,44,61, the auditory cortex as it receives direct efferent projection from S1BF45,61, and the VPL nuclei of the thalamus connected via corticothalamic projections45.“

- *It would be interesting to see bilateral traces as supplementary figures.*

We have provided the Supplementary Figure 5 that contains both ipsilateral and contralateral responses to whiskers stimulation (from both left and right pads) for all trials and all rats included in this work.

- *In both panels C and D, n=5 is reported, but methods state the use of 7 animals. Please clarify how animals have been used in the different studies*

We have clarified the report on animal use and amended the Supplementary Table 1 accordingly.

- *In Panel D, the 95% CI intervals seem particularly narrow. Might this be the result of considering multiple trials as independent events? A GLM analysis would avoid this statistical fallacy.*

We have provided the Supplementary Figure 5 that contains both ipsilateral and contralateral responses to whiskers stimulation (from both left and right pads) for all trials and all rats included in this work. The statistical analysis has been adjusted (see Materials and Methods) and completed with a Supplementary Figure 4

Figure 4

- *See comments above for Figure 3*

We have adjusted the Figure 3 accordingly to reviewer's suggestions

Reviewer #3 (Recommendations For The Authors):

1. *Introduction: Given the emphasis on the awake state, it would be helpful to note that a significant portion of strokes occur during sleep - as well as comment on its hemodynamic difference with respect to an awake state.*

We agree with the reviewer on the remark that some strokes occur during sleep phase. However, here the awake state, which has been poorly addressed in the literature, is opposed to anesthesia a condition largely used to investigate brain functions after stroke. We added a point and corresponding references about wake-up stroke, see Line 49.

1. *The effects of anesthetics on stroke are quite variable and the literature data on the topic is rather divergent: it would be helpful for the introduction to reflect the large level of discord in the literature and the wide-ranging mechanisms of action of different anesthetics.*

We thank the reviewer for this comment. We have completed our original sentence in the introduction to better reflect the various effects of anesthetics on stroke, see Line 50

1. *The reference list (14-17) to other studies of brain hemodynamic changes post ischemic stroke is egregiously short. Please expand. Similarly, the list of citations to other functional ultrasound rodent studies in the literature (23-24) is misleading: other groups have published similar work and ought to be cited.*

We thank the reviewer for this comment and added complementary references. However, we believe that the references 14-17 pointed by the reviewer are not only referring to brain hemodynamic changes but mostly on network and function as stated in the manuscript. Regarding references on fUS (23-24) mentioned by the reviewer, we did not limit our citation on functional ultrasound imaging to those 2 articles but on 15+ from 4 different research groups.

1. *It would be helpful if the authors used "spreading depolarization" the way it has been utilized in the many decades of research on them in the literature, namely, as waves of hyper/hypoactivity in the electrophysiological signals. Please use a distinct term to refer to waves of changes in the hemodynamic state.*

We have amended the terminology used in the manuscript. "Spreading depolarization" has been replaced by "hemodynamic events associated with spreading depolarizations" or similar.

1. *Why is this investigation restricted to male rats?*

As a proof of concept, we did not performed experiments in female rats. We agree that further investigation would require a gender mix. We added a line in the discussion.

New text – Line 455: "Finally, it is important to note that this proof-of-concept work did not specifically focus the impact of sex dimorphism on the stroke or early behavioral outcomes following the insult that would greatly enhance the translational value of such preclinical stroke study⁸⁰."

1. *Were the animals tested during their active phase? If not, why not, and what are the implications of testing their responses during the sleep phase?*

We think there is a misunderstanding here as we investigated brain functions in awake head-fixed rats. Therefore, the sleep/active phases were not investigated neither mentioned in the manuscript.

1. *How is the level of stress monitored/established?*

In this work, we followed established procedure used to reduce stress and discomfort of the rats all along the experiment. The procedure used is now better detailed in the Materials and Methods section. However, the level of stress was not monitored, and would be of interest to consider in future experiments.

1. *What are the sequelae of stress on brain hemodynamics, especially given 1-4 hour long sessions.*

This is a good remark. While we cannot state on how the stress impacts brain hemodynamics, the data extracted show that hemodynamics response functions were stable and robust over hour-long recording (see control and pre-stroke sessions in Supplementary Figure 5).

1. *How is the animal prepared for stroke induction? In general, the methodological steps surrounding animal handling and preparation are exceedingly terse.*

We provided more details about the handling and preparation of the rats in the Materials and Methods section.

Original text: "Body restraint and head fixation.

Rats were habituated to the workbench and to be restrained in a sling suit (Lomir Biomedical inc, Canada), progressively increasing the restraining period from minutes to hours^{33,34}. After the headpost implantation (see below), rats were habituated to be head-fixed while restrained in the sling. The period of fixation was progressively increased from minutes to

hours. Water and food gel (DietGel, ClearH2O, USA) were provided along the habituation session. Once habituated, the cranial window for imaging was performed as described below (Figure 1A-C)."

New text - Line 90: "Body restraint and head fixation.

The body restraint and head fixation procedures are adapted from published protocols and setup dedicated for brain imaging of awake rats^{39–41}. Rats were habituated to the workbench and to be restrained in a sling suit (Lomir Biomedical inc, Canada) by progressively increasing restraining periods from minutes (5mins, 10mins, 30mins) to hours (1 and 3hrs) for one or two weeks. The habituation to head-fixation started by short (5 to 30s) and gentle head-fixation of the headpost between fingers. The headpost was then secured between clamps for fixation periods progressively increased following the same procedure as with the sling. For both body restraint and head fixation, the initial struggling and vocalization diminished over sessions. Water and food gel (DietGel, ClearH2O, USA) were provided for all body restraint and head-fixation habituation sessions. Once habituated, the cranial window for imaging was performed as described below (Figure 1A-C)."

| 1. *What is the reproducibility of the chemo-thrombotic model timeline? What are its limitations?*

We have provided more information on the chemo-thrombotic model and its limitations in the discussion section to discuss

New text – Line 402: "However, to adequately and efficiently occlude the vessel of interest, removing a piece of skull remains required. As mentioned in the report on animal use, one rat was excluded from the analysis as the MCA spontaneously reperuses, thus dropping the success rate of such model."

| 1. *What is the motivation behind the 5-days post stroke timepoint selection?*

In addition to demonstrating the feasibility of imaging brain functions at different timepoint following the ischemia, the motivation to performed this delayed session was to capture functional diaschisis which is known to occur few days after the initial insult. More recurrent imaging sessions covering a longer post-stroke period would be of high interest to better capture the impact of ischemia on both the brain hemodynamics and functions.

| 1. *How predictive is hyperacute hemodynamics imaging of the long-term outcome?*

We thanks the reviewer for this question, that remains of major interest in the stroke realm. However, the prediction of long-term outcome would require to capture brain hemodynamic at larger scale as performed in Hingot et al., Theranostics 2020 and Brunner et al. JCBFM 2023, a coverage not accessible with the imaging window proposed in this work.

| 1. *It would be greatly reassuring if the authors presented the statistical parametric maps without masking regions of interest (eg Fig3B).*

We thank the reviewer for pointing out this potential confusion. In the first version of the figure, the colormap used of activity maps was indeed non optimal. Therefore, we i) adjusted the colormap used in Fig 3 and 4 and ii) provided non-thresholded z-score maps for all rats in Supplementary Figure 5.

| 1. *Fig 3C is hard to make out.*

We provided a full page version of the Figure 3C in Supplementary Figure 3.

1. *Figs 3,4 should incorporate box and whisker plots of data across all rats scatter plots of individual animal data.*

We are not sure which kind of data the reviewer wants to have displayed here. However, we have provided the Supplementary Figure 5 that contains both ipsilateral and contralateral responses to whiskers stimulation (from both left and right pads) for all trials and for individual animal included in this work.

1. *The final panels in Figures 3,4 would more tellingly include the plots of the linear models fitted.*

Based on all reviewers' comments, we have adjusted and clarified the statistical analysis performed (see Materials and Method) and completed with a Supplementary Figure 4.

1. *The frame rate calculations are not adding up unless averaging and pauses are included so some more details should be stated. Are tilted plane waves averaged before compounding as in prior publications?*

The angles are averaged 6 times before compounding to reduce signal to noise ration and there is a pause of 0.3s between each Doppler image. See also question "Functional Ultrasound Imaging acquisition" from reviewer 2. We also provided supplementary and key information about the sequence used in this work.

We have provided complementary information in the manuscript:

Original text:" The ultrasound sequence generated by the software is the same as in Macé et al.,²⁶ and Brunner, Grillet et al., Briefly, the ultrafast scanner images the brain 140 with 5 tilted plane-waves (-6°, -3°, +0.5°, +3°, +6°) at a 10-kHz frame rate. The 5 plane-wave images are added to create compound images at a frame rate of 500Hz. Each set of 250 compound images is 142 filtered to extract the blood signal. Finally, the intensity of the filtered images is averaged to obtain a 143 vascular image of the rat brain at a frame rate of 1.25Hz. Then, the acquired images are processed with a dedicated GPU architecture, displayed in real-time for data visualization, and stored for subsequent off-line analysis."

New text – Line 146:" The ultrasound sequence generated by the software is adapted from Macé et al.³¹ and Brunner, Grillet et al.³⁴ Ultrafast images of the brain were generated using 5 tilted plane-waves (-6°, -3°, +0.5°, +3°, +6°). Each plane wave is repeated 6 times and the recorded echoes are averaged to increase the signal to noise ration. The 5 plane-wave images are added to create compound images at a frame rate of 500Hz. To obtain a single vascular image we acquired a set of 250 compound images in 0.5s, an extra 0.3s pause is included between each image to have some processing time to display the images for real-time monitoring of the experiment. The set of 250 compound images has a mixed information of blood and tissue signal. To extract the blood signal we apply a low pass filter (cut off 15Hz) and an SVD filter that eliminates 20 singular values. This filter aims to select all the signal from blood moving with an axial velocity higher than ~1mm/s. To obtain a vascular image we compute the intensity of the blood signal i.e., Power Doppler image. This image is in first approximation proportional to the cerebral blood volume^{26,28}. Overall, this process enables a continuous acquisition of power Doppler images at a frame rate of 1.25Hz during several hours."

1. *Ultrasound data processing: The filtering process should have more description. It would be highly instructive to explain that the power Doppler signal is being used and comment clearly on its relationship to blood volume, commenting on stalled flow microvessels/RBC-devoid microvessels, and considerations of vessel orientation.*

The compound image has a mixed information of blood and tissue signal. To extract the blood signal, we applied a low pass filter (cut off 15Hz) and an SVD filter that eliminates 20 singular values. This filter selects all the signal from blood moving with an axial velocity higher than ~1mm/s. To obtain a vascular image we compute the intensity of the blood signal (Power Doppler image). This power Doppler image is in first approximation proportional to the cerebral blood volume.

These information have been added in the Materials and Methods section of the manuscript.

1. *Does the SVD processing have the same cut off (20 singular values) as in prior publications as a standard value, or is that adjusted for each study? There are enough minor differences between sequences that these details are uncertain. Do the overall hemodynamics measurements (Fig 2) include all data acquired, or do they exclude the whisker stimulation events, and if so, how long of a window is excluded? The explanation of the activity maps should be rephrased e.g. "... recordings are segmented in shorter 40-s time windows encompassing the whisker stimulation trials..."*

We agree that these details are important, all these information have been added to the manuscript

- SVD processing: We eliminate 20 singular values as in cited studies.
- Sequence: we have included more details about the sequence.
- Processing: all data during the whisker stimulation is used.
- We have rephrased the explanation about the activity maps.

1. *Discuss the methodology behind histological data shown in Fig. 1.*

We thank the review for highlighting this omission. We have provided a paragraph in the Materials & Methods section detailing the histology procedure (Line 228):

“Histopathology

Rats were killed 24hrs after the occlusion for histological analysis of the infarcted tissue. Rats received a lethal injection of pentobarbital (100mg/kg i.p. Doletal, Vetoquinol, France). Using a peristaltic pump, they were transcardially perfused with phosphate-buffered saline followed by 4% paraformaldehyde (Sigma-Aldrich, USA). Brains were collected and post-fixed overnight. 50-μm thick coronal brain sections across the MCA territory were sliced on a vibratome (VT1000S, Leica Microsystems, Germany) and analyzed using the cresyl violet (Electron Microscopy Sciences, USA) staining procedure (see Open Lab Book for procedure). Slices were mounted with DPX mounting medium (Sigma-Aldrich, USA) and scanned using a bright-field microscope